



TRINITY GRAMMAR SCHOOL

Mathematics Department

2012

HALF YEARLY EXAMINATION

HSC Assessment Task 3

Year 12

Mathematics

Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided on the back of the Section I answer sheet and on page 15
- Show all necessary working in Questions 11 – 16
- Write your Board of Studies Student Number **and** Class Teacher on the writing booklet(s) **or** sheet(s) submitted
- **WEIGHTING:** 30%

Board of Studies Student Number

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Class Teacher:

Do NOT write solutions on this question paper. Any working on the question paper will NOT be marked.

Total marks – 100

Section I

Pages 3 – 6

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II

Pages 7 – 14

90 marks

- Attempt Questions 11 – 16
- Allow about 2 hours 45 minutes for this section

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Section I 10 marks

- Circle the correct response on the answer sheet provided
 - Each question is worth 1 mark
-

1 Using a suitable substitution, the definite integral $\int_0^{\frac{\pi}{24}} \tan 2x \sec^2 2x \, dx$ is equivalent to

A. $\int_0^{\frac{\pi}{24}} \frac{u}{2} \, du$

B. $\int_0^{2-\sqrt{3}} u \, du$

C. $\int_0^{2-\sqrt{3}} 2u \, du$

D. $\int_0^{2-\sqrt{3}} \frac{u}{2} \, du$

2 In the Argand plane, the curve $|z - (2 + 3i)| = 1$ is intersected exactly twice by the curve with equation

A. $|z - 3i| = 1$

B. $|z - 3i| = |z - 3|$

C. $\operatorname{Im}(z) = 4$

D. $\operatorname{Re}(z) = 3$

3 The slope of the curve $2x^3 - y^2 = 7$ at the point where $y = -3$ is

A. -4

B. -2

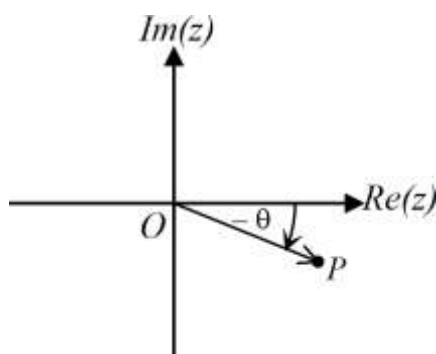
C. 2

D. 4

4 Which one of the following relations does **NOT** have a graph that is a straight line passing through the origin?

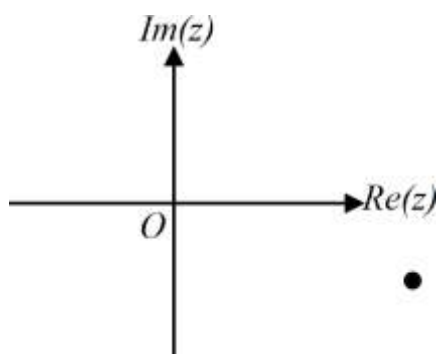
- A. $z + \bar{z} = 0$
- B. $3\operatorname{Re}(z) = \operatorname{Im}(z)$
- C. $z = i\bar{z}$
- D. $\operatorname{Re}(z) + \operatorname{Im}(z) = 1$

5 A certain complex number z , with $|z| > 1$, is represented by the point P on the following Argand diagram below.

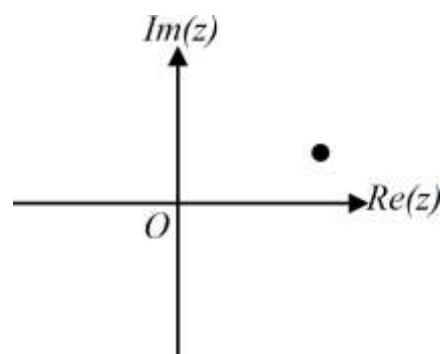


The complex number $\frac{1}{\bar{z}}$ is best represented by

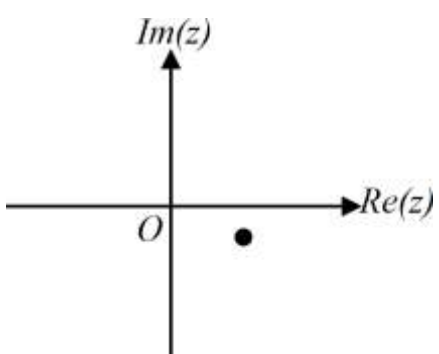
A.



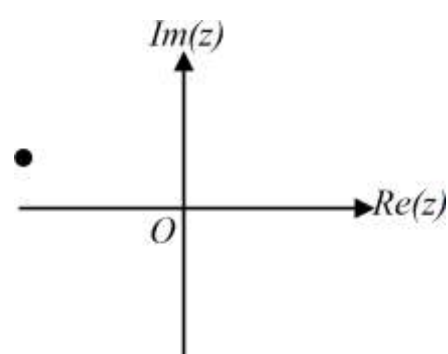
B.



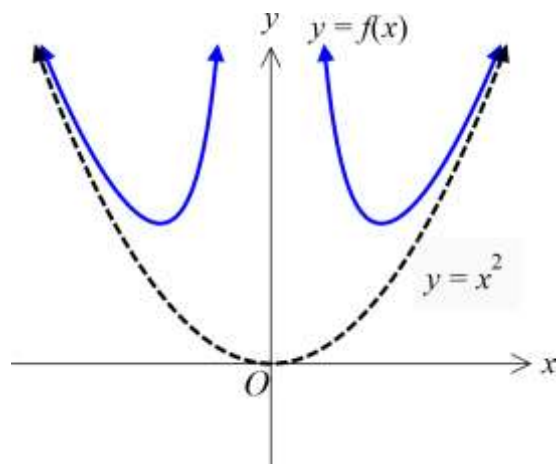
C.



D.



6



A possible equation for the graph of the curve $y = f(x)$ shown above is

A. $y = \frac{x^3 + a}{x}, \quad a > 0$

B. $y = \frac{x^3 + a}{x}, \quad a < 0$

C. $y = \frac{x^4 + a}{x^2}, \quad a > 0$

D. $y = \frac{x^4 + a}{x^2}, \quad a < 0$

7 For a certain complex number z , $\arg(z) = \frac{\pi}{5}$. The complex number z^7 has principal argument of

A. $-\frac{7\pi}{5}$

B. $-\frac{3\pi}{5}$

C. $\frac{3\pi}{5}$

D. $\frac{7\pi}{5}$

-
- 8** Given that $(1+i)^n = ai$, where a is a non-zero real number, then $(1+i)^{2n+2}$ simplifies to
- A. a^4
- B. $2a^2i$
- C. 0
- D. $-2a^2i$
- 9** In simplest form, $\frac{d}{dx} \cos^{-1}(\sin x)$ is equal to
- A. -1 , for all x
- B. -1 , if $\cos x < 0$
- C. 1 , for all x
- D. 1 , if $\cos x < 0$
- 10** Let $z = \cos \theta + i \sin \theta$. The expression $z^n + \frac{1}{z^n}$ is equivalent to
- A. $-2 \cos n\theta$
- B. $2 \cos n\theta$
- C. $-2i \sin n\theta$
- D. $2i \sin n\theta$

End of Section I

Section II 90 marks

- Begin each question in a new writing booklet or on a new answer sheet
 - Show all necessary working
 - Each question is worth 15 marks
-

Question 11 (15 marks)

(a) Find $\operatorname{Im}\left(\frac{3+4i}{1+2i}\right)$. 2

(b) Express $z = i - 1$ in modulus-argument form. 1

(c) Find, in modulus-argument form, all the roots of $z^3 = -8$. 2

(d) Sketch on separate Argand diagrams the locus of a point $z = x + yi$ such that:

(i) $2|z| = z + \bar{z} + 4$ 2

(ii) $\operatorname{Im}(z^2) = -2$ 2

(iii) $|\operatorname{Re}(z)| > 1$ 2

(e) A complex number z satisfies the equations $2|z-1| = |z|$ **and** $\arg(z-1) - \arg z = \frac{\pi}{3}$.

(i) Show that $\frac{z-1}{z} = \frac{1}{2} \operatorname{cis} \frac{\pi}{3}$. 2

(ii) Hence, or otherwise, solve for z . Leave your answer in Cartesian form. 2

Question 12 (15 marks)**Begin a NEW answer booklet or answer sheet**

(a) Using the substitution $u = \ln x$, evaluate $\int_1^{e^3} \frac{(\ln x)^3}{x} dx$. 3

(b) Find $\int \cos^3 x \, dx$. 2

(c) (i) Show that $f(x) = x \sin x$ is an even function about the line $x = 0$. 2

(ii) Find, using integration by parts, the area of the region bounded by $y = x \sin x$, $|x| = \frac{\pi}{2}$ and the x -axis. 3

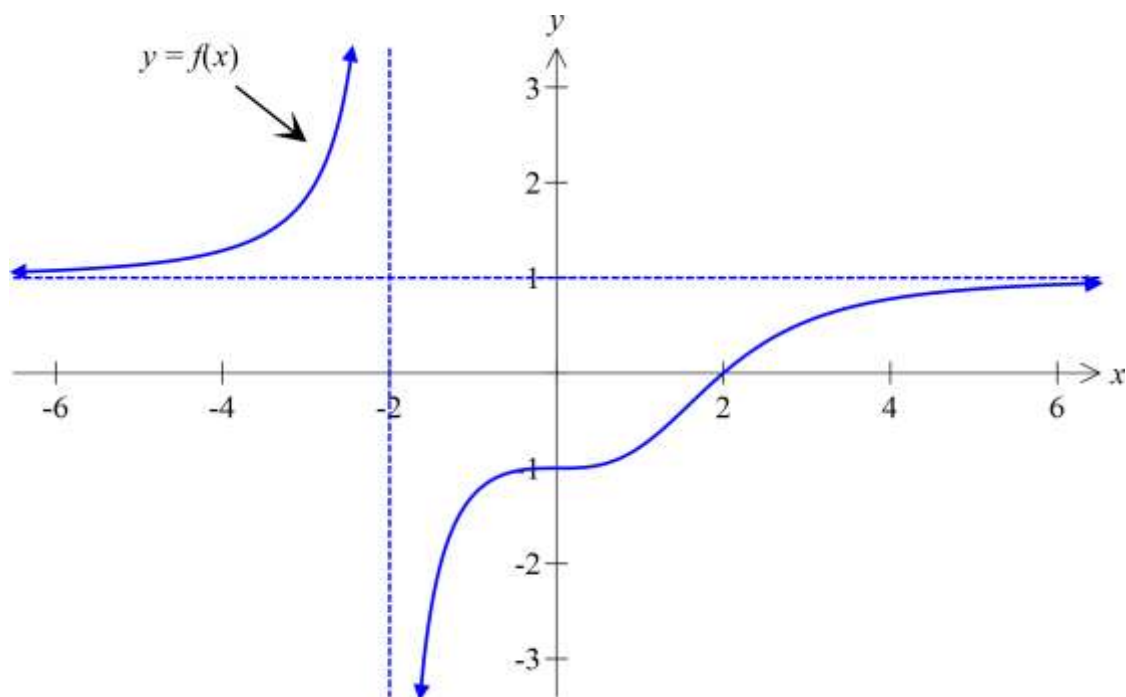
(d) (i) Find real values A , B and C such that: 3

$$\frac{9+x-2x^2}{(1-x)(3+x^2)} = \frac{A}{1-x} + \frac{Bx+C}{3+x^2}$$

(ii) Hence find $\int \frac{9+x-2x^2}{(1-x)(3+x^2)} dx$. 2

Question 13 (15 marks)

Begin a NEW answer booklet or answer sheet



- (a) The diagram above shows the graph of a function $y = f(x)$.

On the separate answer sheet provided, sketch the graphs of:

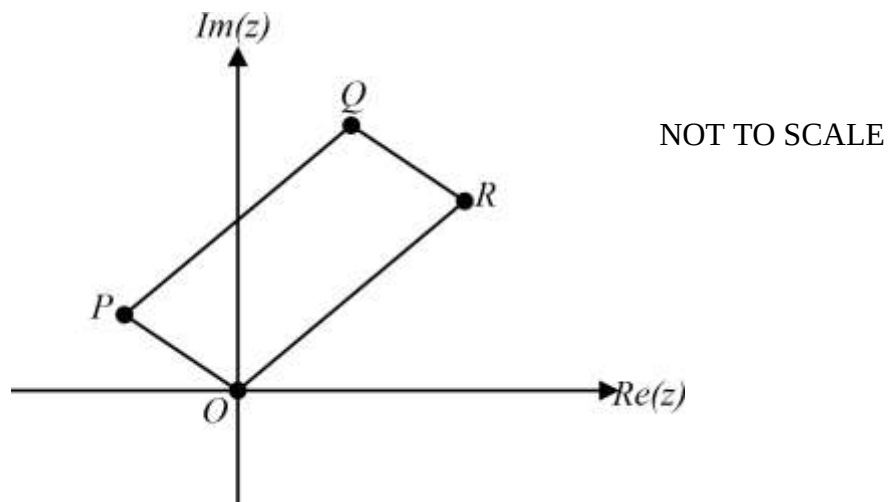
- | | | |
|-------|----------------------|---|
| (i) | $y = \frac{1}{f(x)}$ | 2 |
| (ii) | $y = f(-x)$ | 1 |
| (iii) | $y = f x $ | 1 |
| (iv) | $ y = f(x)$ | 1 |
| (v) | $y^2 = f(x)$ | 2 |
| (vi) | $y = \log_e f(x)$ | 2 |

- (b) Use the principle of mathematical induction to show that $3^n > n^3$, for positive integers $n > 3$. 3

Question 13 continues on the next page...

Question 13 continued:

(c)



In the diagram above, $OPQR$ is a parallelogram with $OP = \frac{1}{2}OR$.

The point P represents the complex number $-\frac{1}{2} + \frac{\sqrt{3}}{2}i$.

If $\angle POR = 60^\circ$, find in Cartesian form, the complex numbers representing

- | | | |
|------|-------|---|
| (i) | R ; | 2 |
| (ii) | Q . | 1 |

Question 14 commences on the next page

Question 14 (15 marks)**Begin a NEW answer booklet or answer sheet**

(a) Let $y = (x^2 - 1)\sqrt{x+1}$.

(i) State the domain of $y = (x^2 - 1)\sqrt{x+1}$. **1**

(ii) Find the x and y -intercept(s). **2**

(iii) Let $\frac{dy}{dx} = \frac{5x^2 + 4x - 1}{2\sqrt{x+1}}$ and $\frac{d^2y}{dx^2} = \frac{3(5x^2 + 8x + 3)}{4(x+1)\sqrt{x+1}}$ (DO NOT PROVE THIS). **3**

Find the coordinate(s) of any stationary point(s) and determine their nature.

(iv) Sketch the graph of $y = (x^2 - 1)\sqrt{x+1}$. **2**

(v) Find the area enclosed by $y = (x^2 - 1)\sqrt{x+1}$ and the x -axis. **4**

Give your final answer in the form $\frac{a\sqrt{b}}{c}$, where a , b and c are non-zero integers.

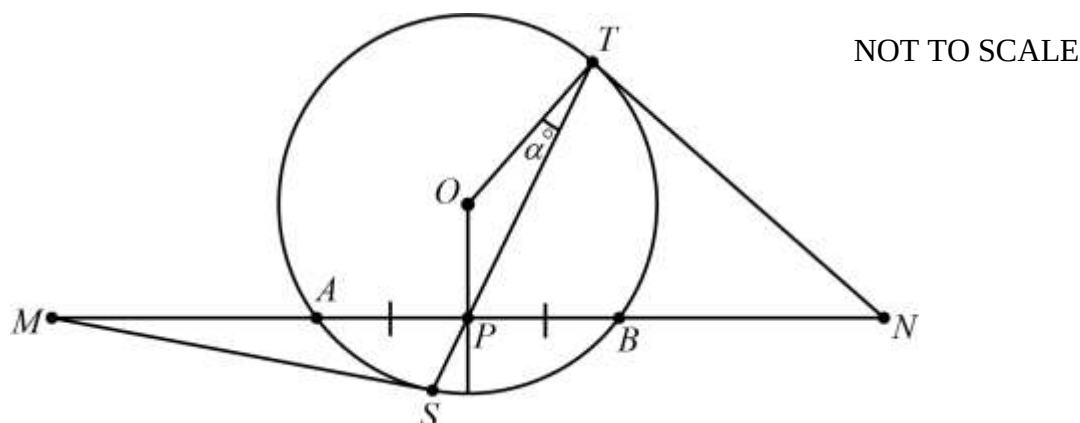
(b) (i) Expand and simplify $\cos(A + B) - \cos(A - B)$. **1**

(ii) Hence or otherwise, solve $\cos x - \cos 3x = 0$. **2**

Give your final answer(s) in general form.

(a) Evaluate $\int_0^{\frac{\pi}{2}} \frac{2}{1 + \cos \theta + \sin \theta} d\theta$, using the substitution $t = \tan \frac{\theta}{2}$. 3

(b)



In the diagram above, P is the midpoint of the chord AB in the circle with centre O . A second chord ST passes through P and the tangents at the endpoints meet AB produced at M and N respectively.

Copy or trace this diagram into your writing booklet.

- (i) Explain why $OPNT$ is a cyclic quadrilateral. 2
- (ii) Explain why $OPSM$ is a cyclic quadrilateral. 2
- (iii) Let $\angle OTS = \alpha$. Show that $\angle ONP = \angle OMP = \alpha$. 2
- (iv) Prove that $AM = BN$. 2

(c) Let $f(x) = 2 \cos^{-1} \frac{x}{\sqrt{2}} - \sin^{-1}(1 - x^2)$ for $0 \leq x \leq 1$.

- (i) Show that $f'(x) = 0$. 2
- (ii) Hence evaluate $\int_0^1 f(x) dx$. 2

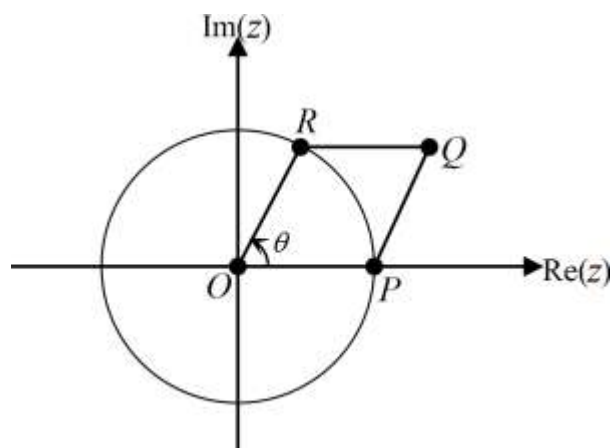
Question 16 (15 marks)**Begin a NEW answer booklet or answer sheet**

- (a) (i) Sketch the graphs of $f(x) = |3x+3| + |x-1|$ and $g(x) = 4x+3$ on the same number plane. **3**
- (ii) Hence, or otherwise, solve for x where $f(x) \leq g(x)$. **2**
- (b) Let $I_n = \int_0^1 x^n e^{-x} dx$, where n is an integer such that $n \geq 0$.
- (i) Evaluate I_0 . **1**
- (ii) Show that $I_n = nI_{n-1} - \frac{1}{e}$, for $n > 0$. **2**
- (iii) Hence, evaluate I_4 . **2**

Question 16 continues on the next page...

Question 16 continued:

(c)



In the diagram above, R represents the complex number $z_1 = \cos \theta + i \sin \theta$.
 P represents the complex number $1 + 0i$ and Q represents the complex number $z_2 = 1 + z_1$. Quadrilateral $OPQR$ is a rhombus.

(i) Show that $z_2 = 2 \cos \frac{\theta}{2} \left(\cos \frac{\theta}{2} + i \sin \frac{\theta}{2} \right)$. **3**

(ii) Hence, show that $\frac{1}{z_2} = \frac{1}{2} - \frac{i}{2} \tan \frac{\theta}{2}$. **2**

End of Section II

End of Examination

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(+\sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(+\sqrt{x^2 + a^2} \right)$$

Note $\ln x = \log_e x, \quad x > 0$

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

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$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(+\sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(+\sqrt{x^2 + a^2} \right)$$

Note $\ln x = \log_e x, \quad x > 0$



TRINITY GRAMMAR SCHOOL
2012, Year 12 Mathematics Extension 2
Half Yearly Examination, HSC Assessment Task 3
SECTION I
ANSWER SHEET

Name:

Class Teacher:

Be sure to write your answers for **Section I** on this answer sheet. After you have selected an answer, **CIRCLE** the correct answer. To change an answer, erase your previous mark completely, and then record your new answer. **Mark only one answer for each question.**

- | | | | | |
|------|---|---|---|---|
| Q1. | A | B | C | D |
| Q2. | A | B | C | D |
| Q3. | A | B | C | D |
| Q4. | A | B | C | D |
| Q5. | A | B | C | D |
| Q6. | A | B | C | D |
| Q7. | A | B | C | D |
| Q8. | A | B | C | D |
| Q9. | A | B | C | D |
| Q10. | A | B | C | D |

A table of Standard Integrals is printed on the reverse side



TRINITY GRAMMAR SCHOOL
2012, Year 12 Mathematics Extension 2
Half Yearly Examination, HSC Assessment Task 3
SECTION II
QUESTION 13 (a) ANSWER SHEET

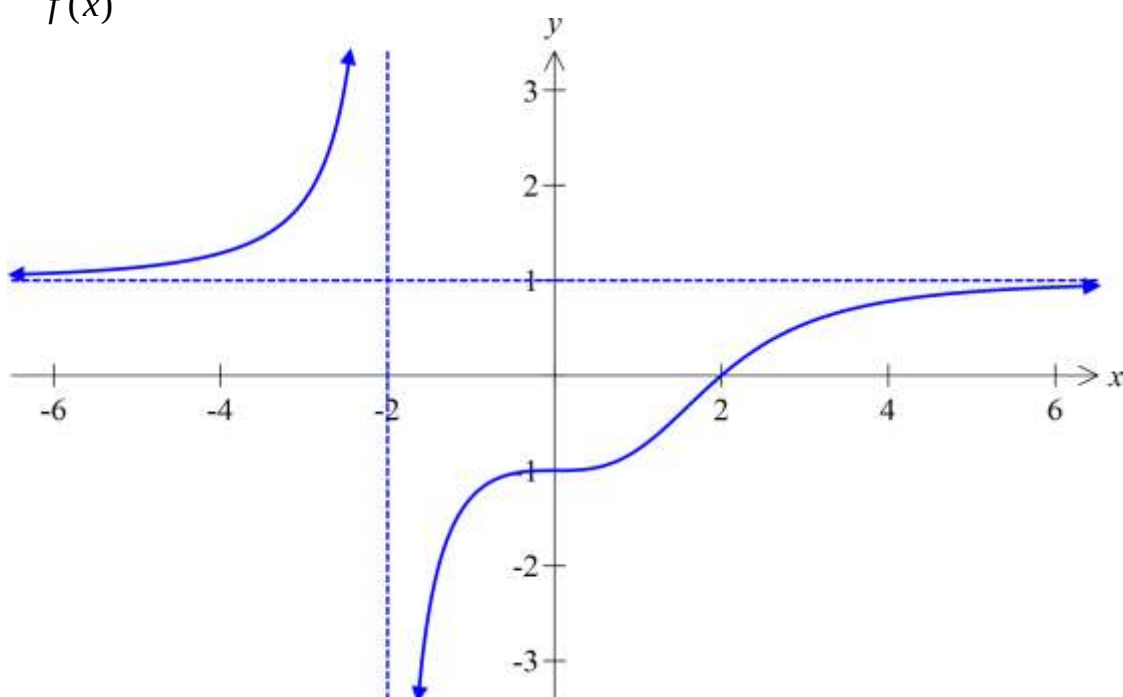
BOS Student Number:

Class Teacher:

Each diagram below shows the graph of $y = f(x)$. On the number planes supplied below, sketch:

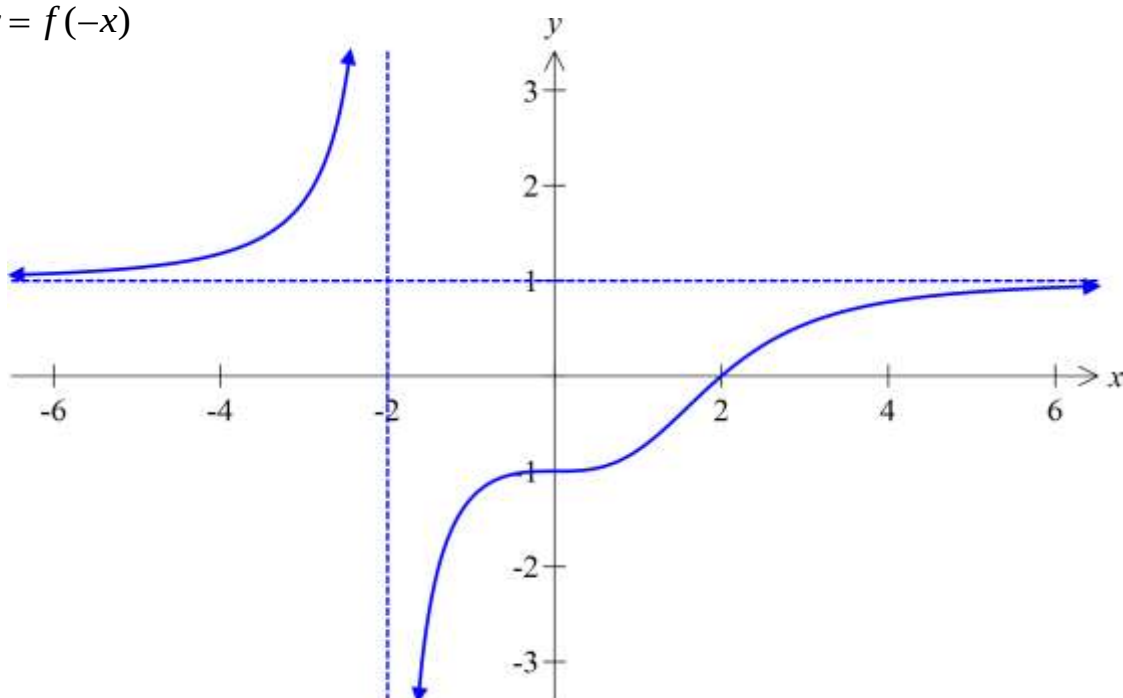
(i) $y = \frac{1}{f(x)}$

2



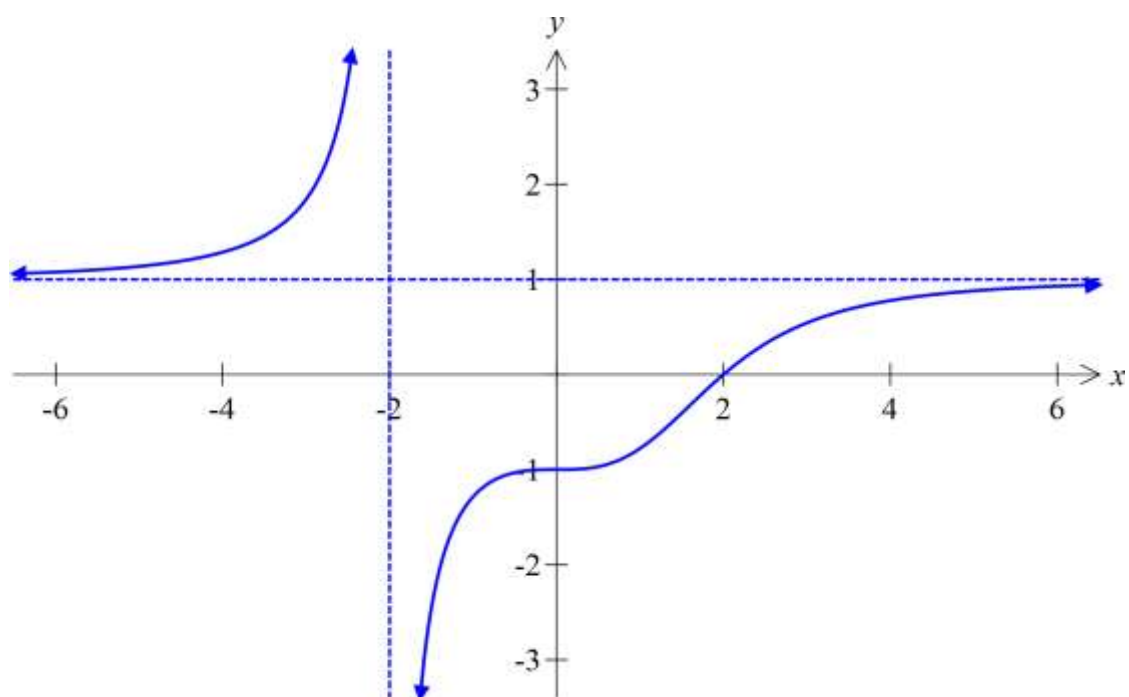
(ii) $y = f(-x)$

1



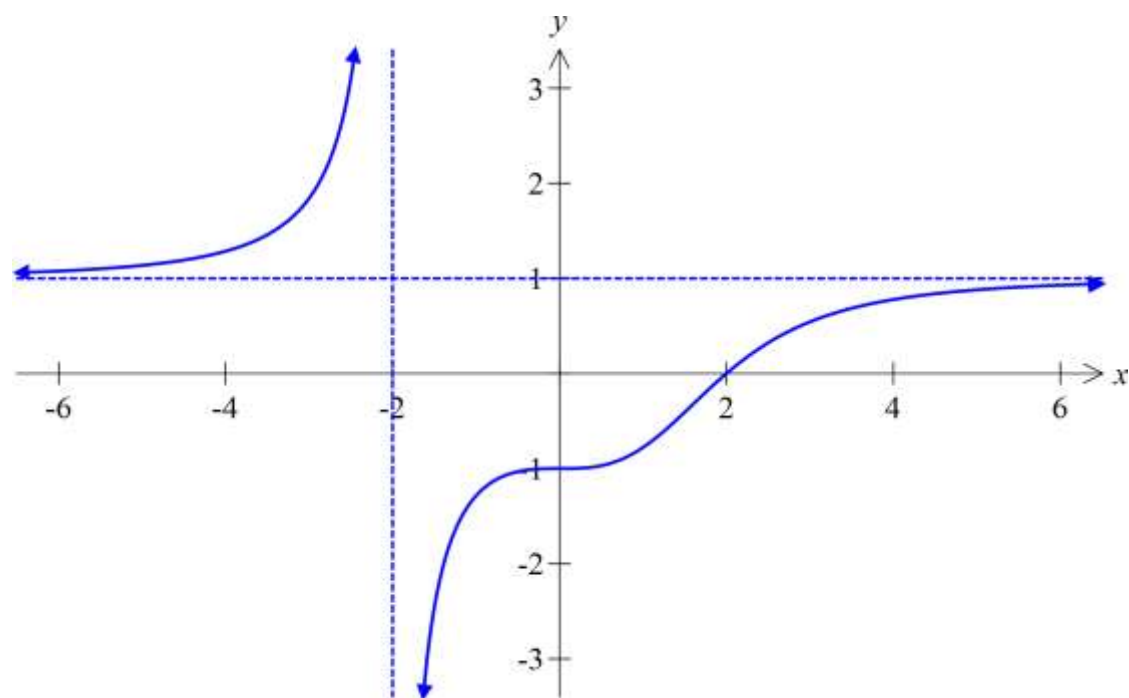
(iii) $y = f |x|$

1



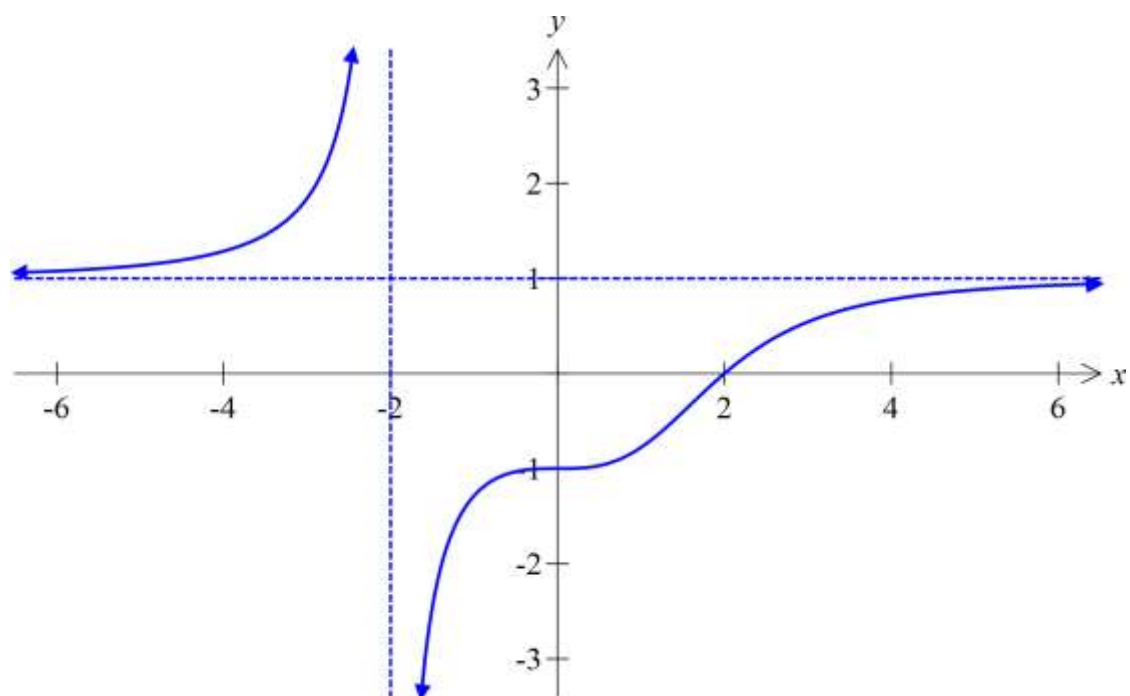
(iv) $|y| = f(x)$

1



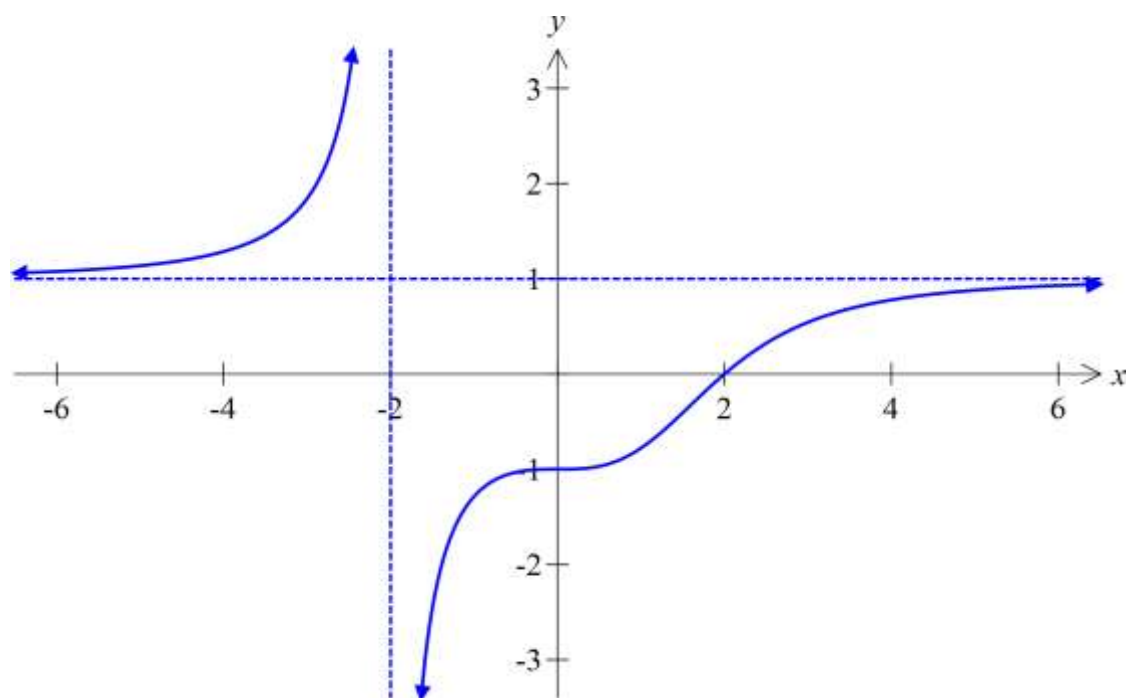
(v) $y^2 = f(x)$

2



(vi) $y = \log_e f(x)$

2



End of Section II, Question 13(a)



TRINITY GRAMMAR SCHOOL
2012, Year 12 Mathematics Extension 2
Half Yearly Examination, HSC Assessment Task 3
SECTION II
QUESTION 13 (a) ANSWER SHEET

ANSWERS

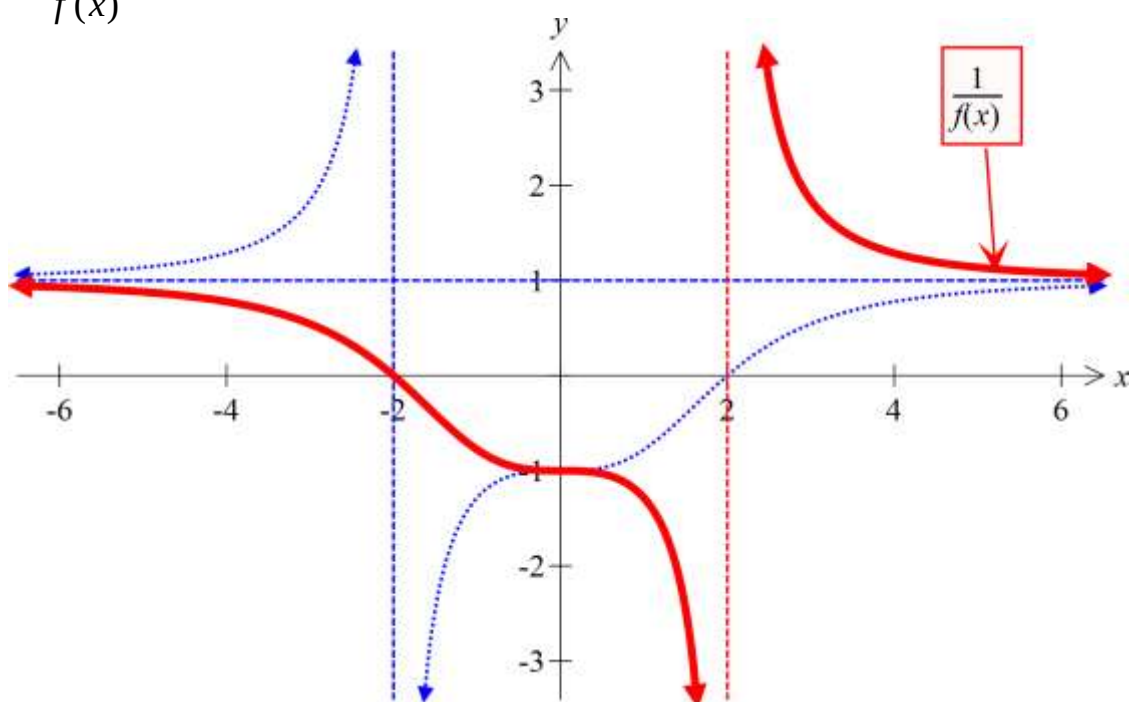
BOS Student Number:

Class Teacher:

Each diagram below shows the graph of $y = f(x)$. On the number planes supplied below, sketch:

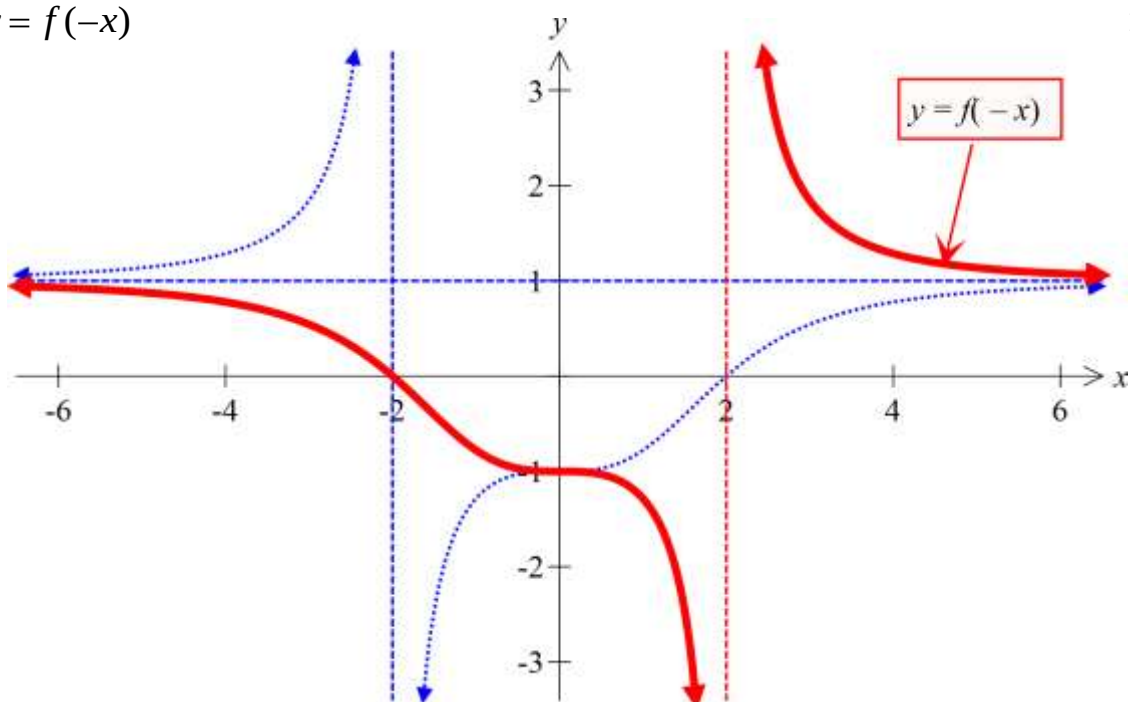
(i) $y = \frac{1}{f(x)}$

2



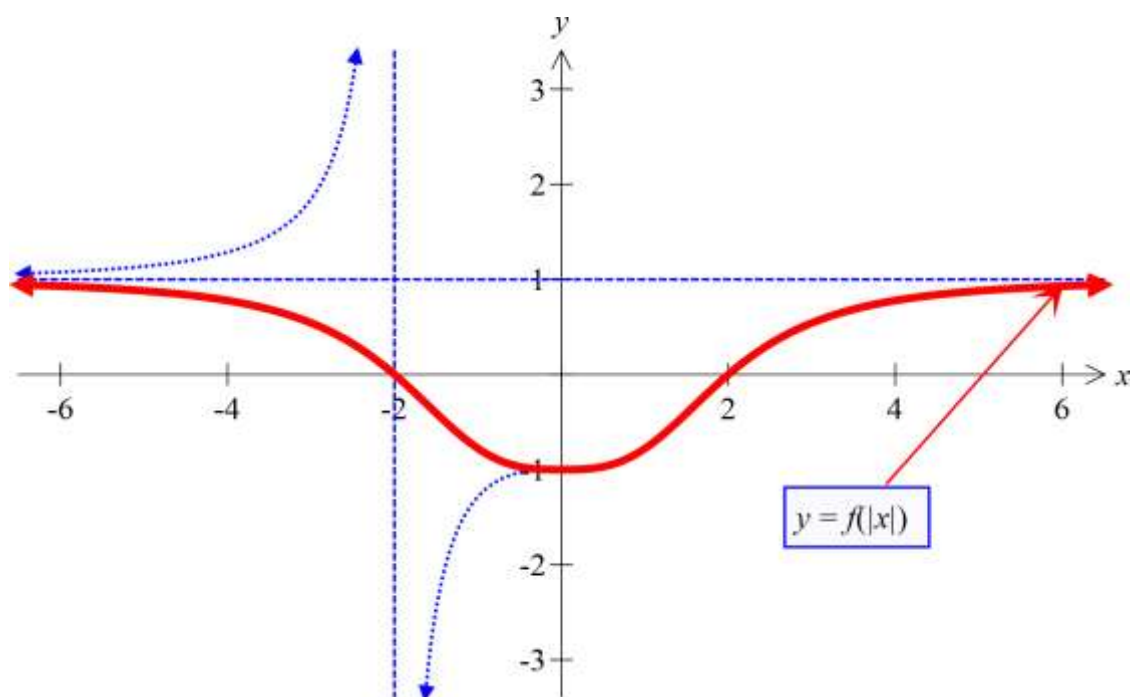
(ii) $y = f(-x)$

1



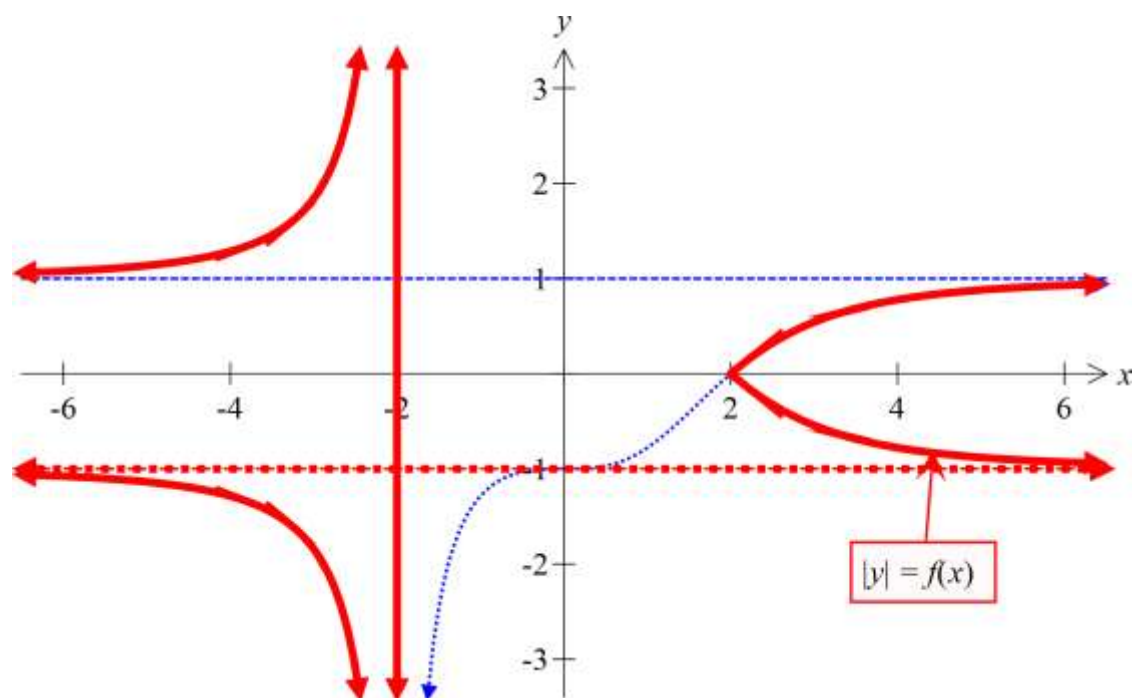
(iii) $y = f(|x|)$

1



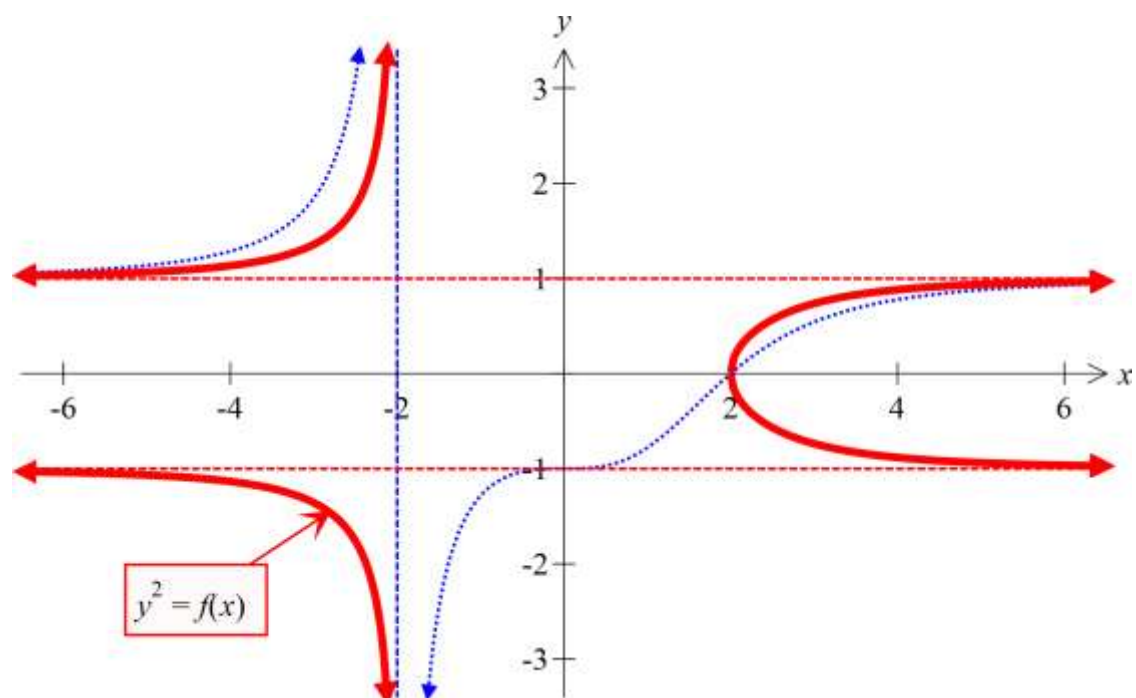
(iv) $|y| = f(x)$

1



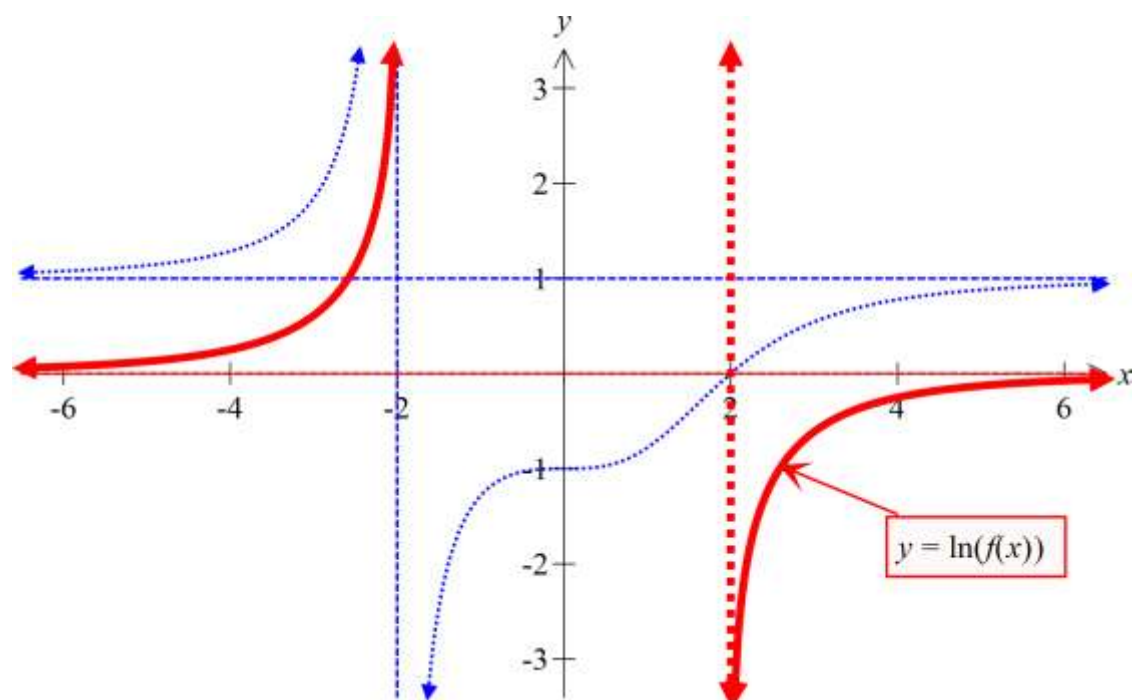
(v) $y^2 = f(x)$

2



(vi) $y = \log_e f(x)$

2



End of Section II, Question 13(a)

3.	Using implicit differentiation, $6x^2 - 2y \frac{dy}{dx} = 0$. Therefore $\frac{dy}{dx} = \frac{3x^2}{y}$. When $y = -3$, $2x^3 - 9 = 7$ i.e. $x^3 = 8 \Rightarrow x = 2$ $\therefore \frac{dy}{dx} = \frac{3(2)^2}{-3} = -4$ Answer: A	1
4.	A. $\Rightarrow 2x = 0$ B. $\Rightarrow 3x = y$ C. $\Rightarrow x + yi = i(x - yi)$ $x + yi = ix + y$ $y(i - 1) = x(i - 1)$ $\therefore y = x$ D. $x + y = 1$ Answer: D	1
5.	$z > 1 \Rightarrow \text{mod}(z) > 1$ $z = z \text{cis}(-\theta)$ $\therefore \bar{z} = z \text{cis}(\theta)$ $\Rightarrow \frac{1}{\bar{z}} = \frac{1}{ z \text{cis}(\theta)}$ i.e. $\left \frac{1}{\bar{z}} \right = \frac{1}{ z }$ $\arg\left(\frac{1}{\bar{z}}\right) = \arg 1 - \arg \bar{z} = 0 - \theta = -\theta$ $\therefore \frac{1}{\bar{z}} = \frac{1}{ z } \text{cis}(-\theta)$ Answer: C	1
6.	A. $y = \frac{x^3 + a}{x} = x^2 + \frac{a}{x}, a > 0$ B. $y = \frac{x^3 + a}{x} = x^2 + \frac{a}{x}, a < 0$ C. $y = \frac{x^4 + a}{x^2} = x^2 + \frac{a}{x^2}, a > 0$ D. $y = \frac{x^4 + a}{x^2} = x^2 + \frac{a}{x^2}, a < 0$ Answer: C	1

7.	$\arg(z^7) = 7 \arg z = \frac{7\pi}{5}$ $\text{Principal } \arg(z^7) = \frac{7\pi}{5} - 2\pi = -\frac{3\pi}{5}$ Answer: B	1
8.	$(1+i)^{2n+2} = [(1+i)^n]^2 (1+i)^2$ $= [ai]^2 (1+2i-1)$ $= -a^2(2i)$ $= -2a^2i$ Answer: D	1
9.	$\frac{d}{dx} \cos^{-1}(\sin x) = -\frac{\cos x}{\sqrt{1-\sin^2 x}}$ $= -\frac{\cos x}{ \cos x }$ $= 1, \text{ if } \cos x < 0$ Answer: D	1
10.	$z^n + \frac{1}{z^n} = \operatorname{cis} n\theta + \operatorname{cis}(-n\theta)$ $= \cos n\theta + i \sin n\theta + \cos(-n\theta) + i \sin(-n\theta)$ $= 2 \cos n\theta$ Answer: B	1

D B A D C C B D D B

QUESTION 11
MARKING GUIDELINES

(a)

Suggested solution	CRITERIA/COMMENTS	MARKS
$\operatorname{Im}\left(\frac{3+4i}{1+2i}\right) = \operatorname{Im}\left(\frac{3+4i}{1+2i} \times \frac{1-2i}{1-2i}\right)$ $= \operatorname{Im}\left(\frac{3+4i}{1+2i} \times \frac{1-2i}{1-2i}\right)$ $= \operatorname{Im}\left(\frac{11-2i}{5}\right)$ $= -\frac{2}{5}$	Correct solution	2
	- Obtaining $\frac{11-2i}{5}$ or equivalent form - Attempt to realise the denominator e.g. writing anywhere $\frac{3+4i}{1+2i} \times \frac{1-2i}{1-2i}$	1

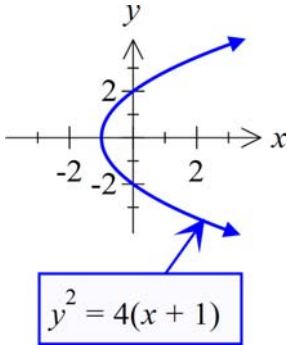
(b)

Suggested solution	CRITERIA/COMMENTS	MARKS
$z = -1 + i = \sqrt{(-1)^2 + (1)^2} \operatorname{cis}\left(\pi - \tan^{-1}\left(\frac{1}{1}\right)\right)$ $= \sqrt{2} \operatorname{cis}\left(\pi - \frac{\pi}{4}\right)$ $= \sqrt{2} \operatorname{cis} \frac{3\pi}{4}$	Correct answer	1

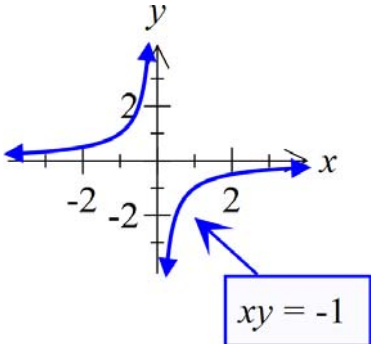
(c)

Suggested solution	CRITERIA/COMMENTS	MARKS
$z^3 = -8 \operatorname{cis} \pi$ $\therefore z = (-8)^{\frac{1}{3}} \operatorname{cis}\left(\frac{\pi + 2k\pi}{3}\right)$ $\text{i.e. } z_1 = -2 \operatorname{cis} \pi$ $z_2 = -2 \operatorname{cis}\left(-\frac{\pi}{3}\right)$ $z_3 = -2 \operatorname{cis}\left(\frac{\pi}{3}\right)$	Correct solution with three correctly listed roots in mod-argument form	2
	- Correct solution with three correct roots in Cartesian form - At least one correct root in Cartesian or mod-arg form - Expressing -8 in mod-arg form - Correctly factorising $z^3 + 8$ into one linear and quadratic factor - Correctly factorising $z^3 + 8$ into three linear factors (one real and two complex factors) - Correctly identifying the roots represent points equally spaced on a circle of radius 2, centre (0, 0) e.g. argument between each complex number is given by $\frac{2\pi}{n}$, where $n = 3$, 120°, or $\frac{2\pi}{3}$	1

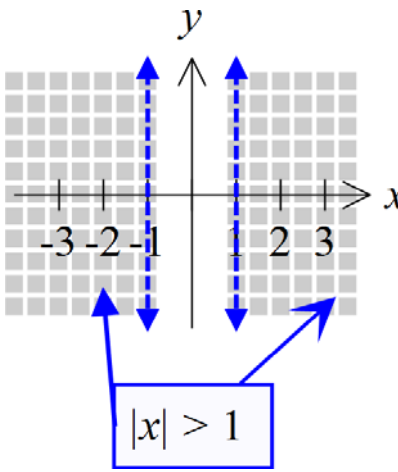
(d) (i)

Suggested solution	CRITERIA/COMMENTS	MARKS
$Let\ z = x + yi$ $2\sqrt{x^2 + y^2} = 2x + 4$ $\sqrt{x^2 + y^2} = x + 2$ $x^2 + y^2 = x^2 + 4x + 4$ <i>i.e.</i> $y^2 = 4(x + 1)$	Correctly positioned (& directed) graph (parabola with vertex at $(-1, 0)$, focus at $(0, 0)$, y-intercepts at $y = 2$ and -2) NB: labelled axes are sufficient	2
	- Attempt to find the Cartesian equation of the locus e.g. replacing z with $x + yi$ into both sides of the given complex equation - Correctly positioned (& directed) parabola but unlabelled axes at key features (such as vertex, focus, intercepts)	1

(d) (ii)

Suggested solution	CRITERIA/COMMENTS	MARKS
$Im(z^2) = -2$ $\Rightarrow Let\ z = x + yi$ $Im(z^2) = 2xy$ $\therefore 2xy = -2$ $xy = -1$	Correctly positioned hyperbola with vertex at $(0, 0)$ and branch in Q2 and Q4, and asymptotes	2
	Attempt to find the Cartesian equation of the locus e.g. replacing z with $x + yi$ into the LHS of the given complex equation	1

(d) (iii)

Suggested solution	CRITERIA/COMMENTS	MARKS
$ \operatorname{Re}(z) > 1$ $\Rightarrow \text{Let } z = x + yi$ $\text{i.e. } x > 1$ 	Correctly positioned 'broken' vertical straight lines through $x = 1$ and $x = -1$ and the correct shaded region	2
	- Attempt to find the Cartesian equation of the locus e.g. replacing z with $x + yi$ into the LHS of the given complex equation - Correctly positioned 'broken' or 'unbroken' vertical straight lines through $x = 1$ and $x = -1$ - A correctly shaded region (that is extending to the right and left of their x-intercept) but axes unlabelled at $x = 1$ or -1 or their intercept. Must have a pair of vertical straight lines though.	1

(e) (i)

Suggested solution	CRITERIA/COMMENTS	MARKS
Given $2 z-1 = z$(1) & $\arg(z-1) - \arg z = \frac{\pi}{3}$(2) From (1) & (2): $\frac{ z-1 }{ z } = \frac{1}{2}$ i.e. $\left \frac{z-1}{z} \right = \frac{1}{2}$ & $\arg\left(\frac{z-1}{z}\right) = \frac{\pi}{3}$ Hence: $\frac{z-1}{z} = \left \frac{z-1}{z} \right \operatorname{cis}\left[\arg\left(\frac{z-1}{z}\right)\right]$ $= \frac{1}{2} \operatorname{cis} \frac{\pi}{3}$	- A correct solution, must refer to the fact (or clearly implied from working) that $\left \frac{z_1}{z_2} \right = \frac{ z_1 }{ z_2 }$ and $\arg(z_1) - \arg(z_2) \pm 2n\pi = \arg\left(\frac{z_1}{z_2}\right)$ - Answer is already given check solution carefully for 'fudged' attempts - A correct solution by changing the LHS to Cartesian form after solving for z in Cartesian or mod-arg form the complex equations & then establishing the required answer(long-winded approach)	2
	- Correctly indicating $\left \frac{z-1}{z} \right = \frac{ z-1 }{ z } = \frac{1}{2}$ - Correctly indicating $\arg\left(\frac{z-1}{z}\right) = \arg(z-1) - \arg(z)$	1

(e) (ii)

Suggested solution	CRITERIA/COMMENTS	MARKS
From(i) $\frac{z-1}{z} = \frac{1}{2} \text{cis} \frac{\pi}{3}$ i.e. $z-1 = z \left(\frac{1}{2} \text{cis} \frac{\pi}{3} \right)$ $z \left(1 - \frac{1}{2} \text{cis} \frac{\pi}{3} \right) = 1$ $z \left(\frac{3}{4} - \frac{\sqrt{3}}{4} i \right) = 1$ $\therefore z = \frac{4}{3 - \sqrt{3}i}$ $= 1 + \frac{\sqrt{3}}{3} i$	Correct solution in $x + yi$ form only	2
	- Correct answer in mod-arg form - Reasonable attempt to find z e.g. showing $z = \frac{4}{3 - \sqrt{3}i}$ or equivalent but not in simplest Cartesian form $x + yi$	1

QUESTION 12 MARKING GUIDELINES

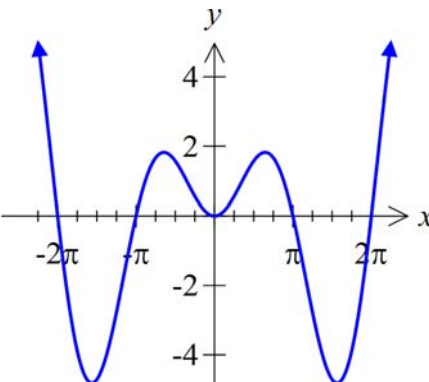
(a)

Suggested solution	CRITERIA/COMMENTS	MARKS
Let $u = \ln x \therefore \frac{du}{dx} = \frac{1}{x}$ when $x = 1, u = \ln 1 = 0$ when $x = e^3, u = \ln e^3 = 3$ $\therefore \int_1^{e^3} \frac{(\ln x)^3}{x} dx = \int_0^3 \frac{(\ln x)^3}{x} \frac{dx}{du} du$ $= \int_0^3 u^3 du$ $= \left[\frac{u^4}{4} \right]_0^3$ $= \frac{81}{4} - 0$ $= \frac{81}{4} \text{ (or } 20\frac{1}{4} \text{ or } 20.25)$	Correct solution using the substitution given (& specified technique) only	3
	- Correct solution using a modified primitive and substitution or equivalent and/or other appropriate method - Correctly finding the new limits (in terms of u) and $\frac{du}{dx}$ and establishing the integrand of u^3 (seen anywhere) but the answer is incorrect.	2
	- Correctly finding the new limits in terms of u - Correctly finding their $\frac{du}{dx}$ - Reasonable attempt to use the substitution technique (any substitution) - Find a correct primitive of their integrand - Correct answer from their incorrect integral statement	1

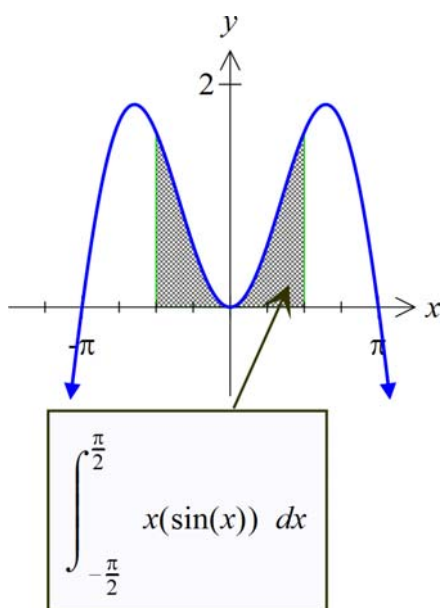
(b)

Suggested solution	CRITERIA/COMMENTS	MARKS
$\int \cos^3 x \, dx = \int \cos^2 x \cos x \, dx$ $= \int (1 - \sin^2 x) \cos x \, dx$ $= \int \cos x - \cos x (\sin x)^2 \, dx$ $= \sin x - \frac{\sin^3 x}{3} + c$ or $\int \cos^3 x \, dx = \frac{1}{4} \left(\frac{\sin 3x}{3} + 3 \sin x \right) + c$	Correct solution in terms of x using any method such as a modified integrand by substitution or integration by parts (+c is not required)	2
	Reasonable attempt to find the primitive of the given expression (+c is not required)	1

(c) (i)

Suggested solution	CRITERIA/COMMENTS	MARKS
$f(x) = x \sin x$ $f(-x) = (-x) \sin(-x)$ $= -x(-\sin x)$ $= x \sin x$ $= f(x)$ Since $f(-x) = f(x)$ then $f(x)$ is an even function 	Algebraically: Correct solution clearly showing a substitution of x with $-x$ and the property $\sin(-x) = -\sin x$ (an odd function) Graphically: A sketch graph of $y = f(x)$ would suffice showing symmetry about $x = 0$.	2
	- Reasonable attempt to sketch the graph of $y = f(x)$ or substituting x with $-x$ in the given expression - Identifying both x and $\sin x$ are odd functions by perhaps drawing their individual graphs and then attempting to use multiplication of ordinates to deduce $y = f(x)$ is an even function	1

(c) (ii)

Suggested solution	CRITERIA/COMMENTS	MARKS
 $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x(\sin(x)) \, dx$ $A = 2 \int_0^{\frac{\pi}{2}} x \sin x \, dx$ Let $u = x \quad dv = \sin x$ $du = 1 \quad v = -\cos x$ $\therefore A = 2 \left[uv - \int v \, du \right]$ $= 2 \left[(-x \cos x) \Big _0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} \cos x \, dx \right]$ $= 2 \left[0 + \int_0^{\frac{\pi}{2}} \cos x \, dx \right]$ $= 2 \int_0^{\frac{\pi}{2}} \cos x \, dx$ $= 2 \times (\sin x) \Big _0^{\frac{\pi}{2}}$ $= 2 \left(\sin \frac{\pi}{2} - \sin 0 \right)$ $= 2 \times 1$ $= 2 \text{ sq. units}$	Correct solution using only the method of integration by parts	3
	Setting up a correct integral statement with correct limits leading to the requested area and uses integration by parts to find a correct primitive but makes an error along the way	2
	- Correct integral statement - Correct primitive based on their integral statement - Correct answer using their incorrect integral or primitive statement - Attempts to use integration by parts e.g. attempts to label u, dv, v, and du seen anywhere	1

(d) (i)

Suggested solution	CRITERIA/COMMENTS	MARKS
$9 + x - 2x^2 \equiv A(3 + x^2) + (Bx + C)(1 - x)$ <i>Let $x = 1$,</i> $\therefore 8 = 4A$ <i>i.e. $A = 2$</i> <i>Let $x = \sqrt{3}i$</i> $\therefore 9 + \sqrt{3}i - 2(-3) = (B\sqrt{3}i + C)(1 - \sqrt{3}i)$ $ = B\sqrt{3}i + 3B + C - C\sqrt{3}i$ <i>i.e. $\begin{cases} 15 = 3B + C \\ 1 = B - C \end{cases}$</i> <i>Solving, $B = 4, C = 3$</i> $\therefore (A, B, C) = (2, 4, 3)$ Note that other values for x can be chosen such as $x = -1$, and $x = 0$.	Correct solution	3
	- Correct values for A, B and C without working - Correct values for any two of A, B or C with supporting working	2
	- A correct value for A, or B or C with/without appropriate working - Writing the statement or equivalent $9 + x - 2x^2 \equiv A(3 + x^2) + (Bx + C)(1 - x)$ - Substituting up to 3 different values of x and attempting to solve their simultaneous equations to find a value for A, B, or C	1

(d) (ii)

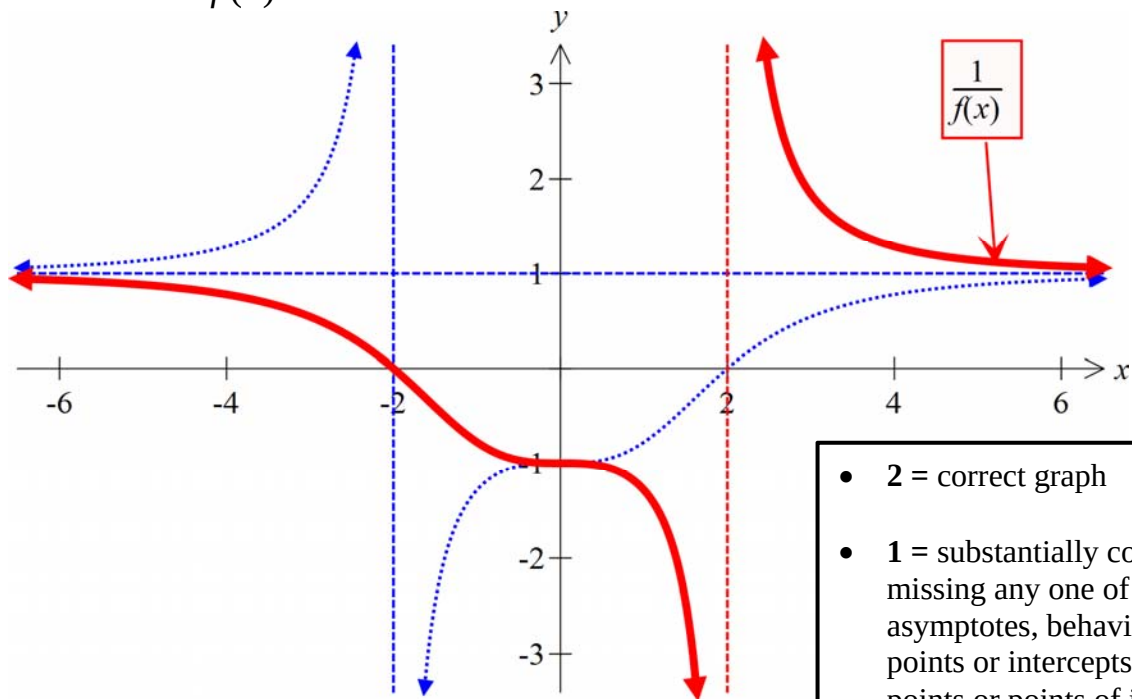
Suggested solution	CRITERIA/COMMENTS	MARKS
$\int \frac{9 + x - 2x^2}{(1 - x)(3 + x^2)} dx = \int \frac{2}{1 - x} + \frac{4x + 3}{3 + x^2} dx$ $= \int \frac{2}{1 - x} + \frac{4x}{3 + x^2} + \frac{3}{3 + x^2} dx$ $= -2 \ln 1 - x + 2 \ln(3 + x^2) + \frac{3}{\sqrt{3}} \tan^{-1} \frac{x}{\sqrt{3}} + c$	Correct solution of their integrand in terms of x using a modified integrand by partial fractions from (i) (provided the integral has not been made easier than intended) (+c is not required)	2
	- Reasonable attempt to find a primitive of the given expression e.g. Splitting the integrand into partial fractions - Correct solution of their integrand in terms of x using any method other than by partial fractions (+c is not required)	1

QUESTION 13 MARKING GUIDELINES

(a)

(i) $y = \frac{1}{f(x)}$

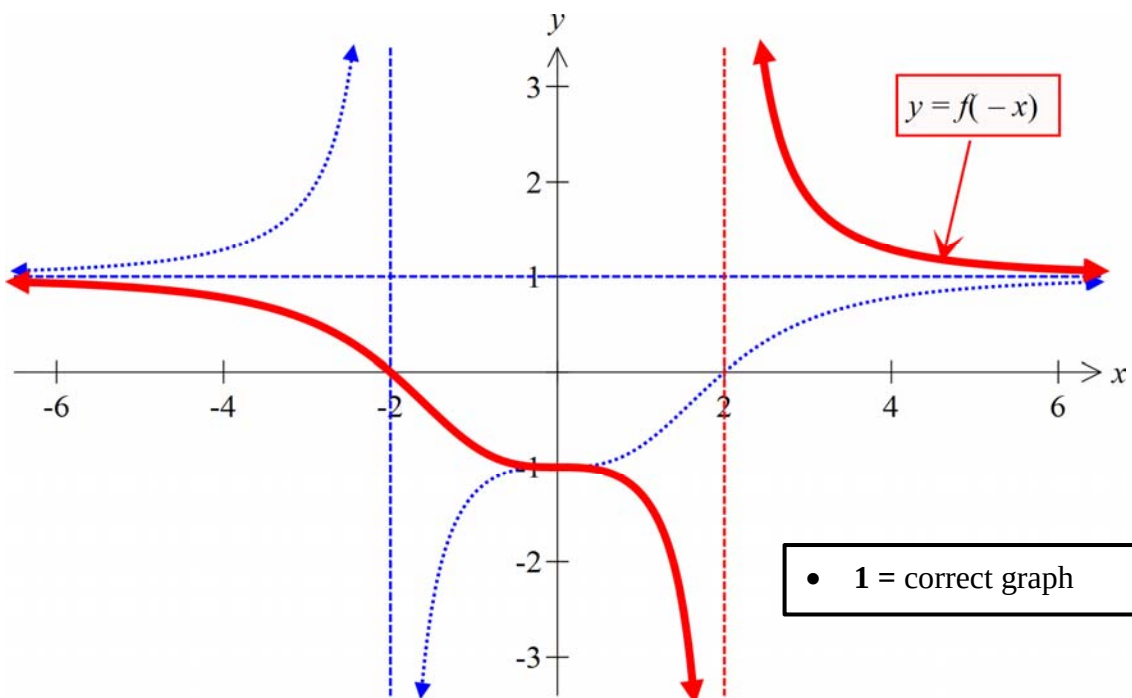
2



- 2 = correct graph
- 1 = substantially correct graph but missing any one of intercepts, asymptotes, behaviour at end points or intercepts, or stationary points or points of inflexion

(ii) $y = f(-x)$

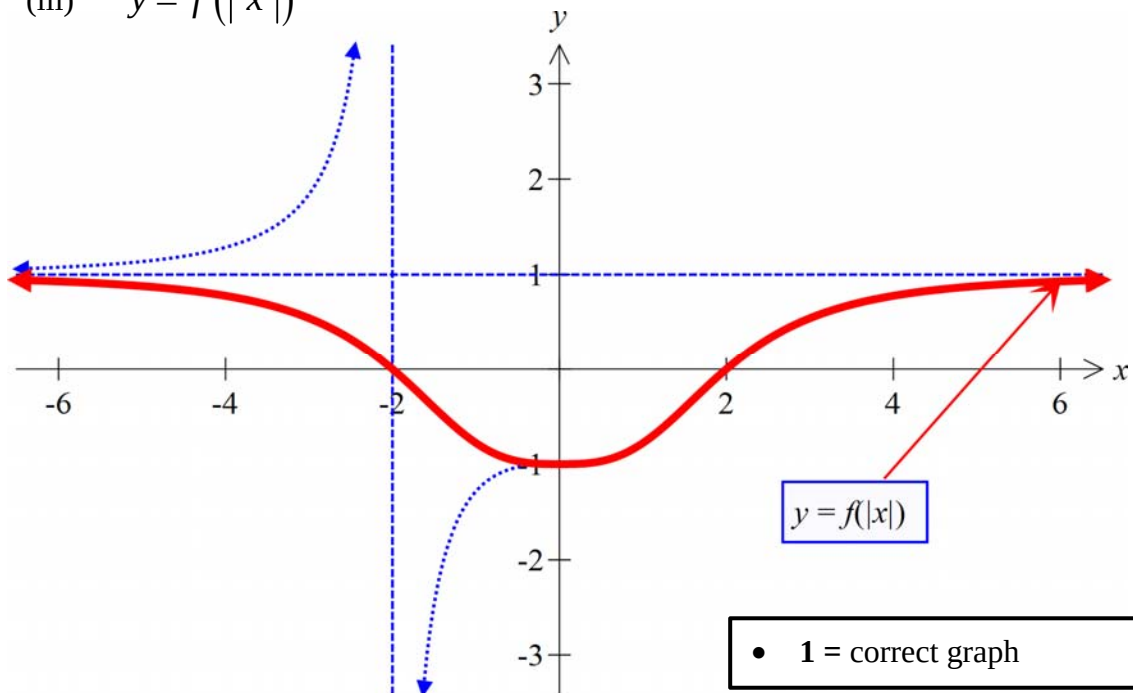
1



- 1 = correct graph

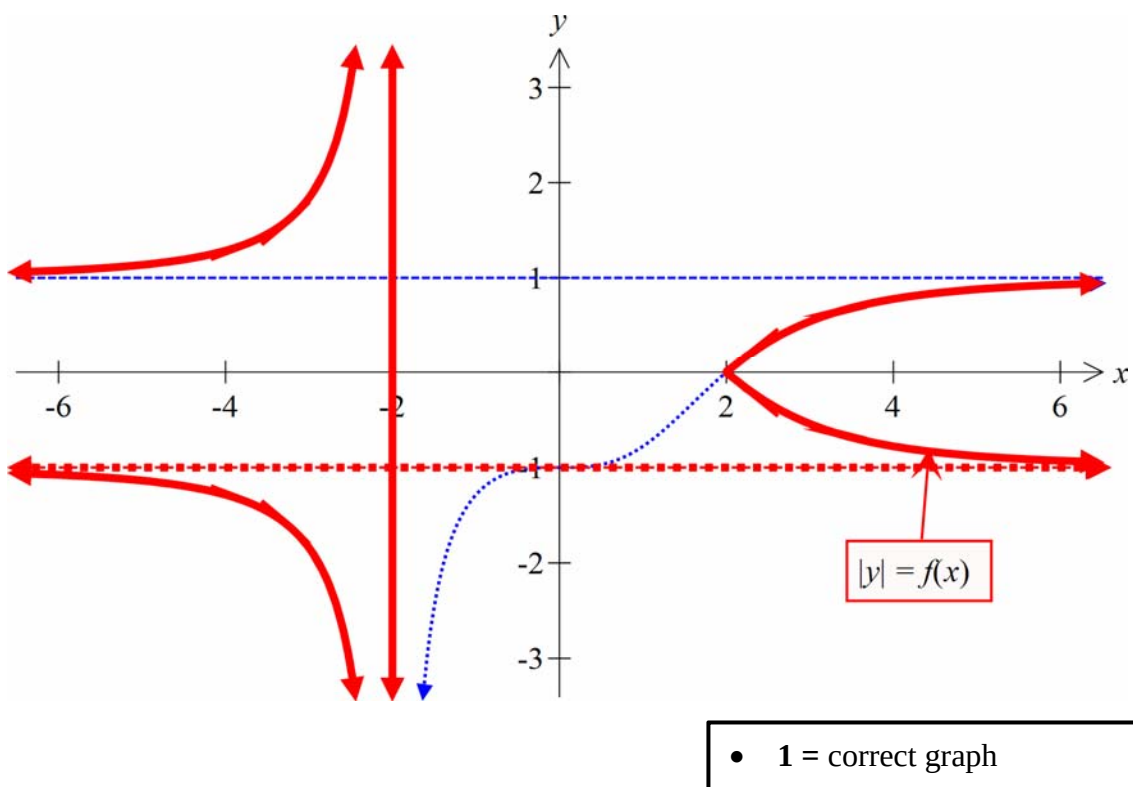
(iii) $y = f(|x|)$

1



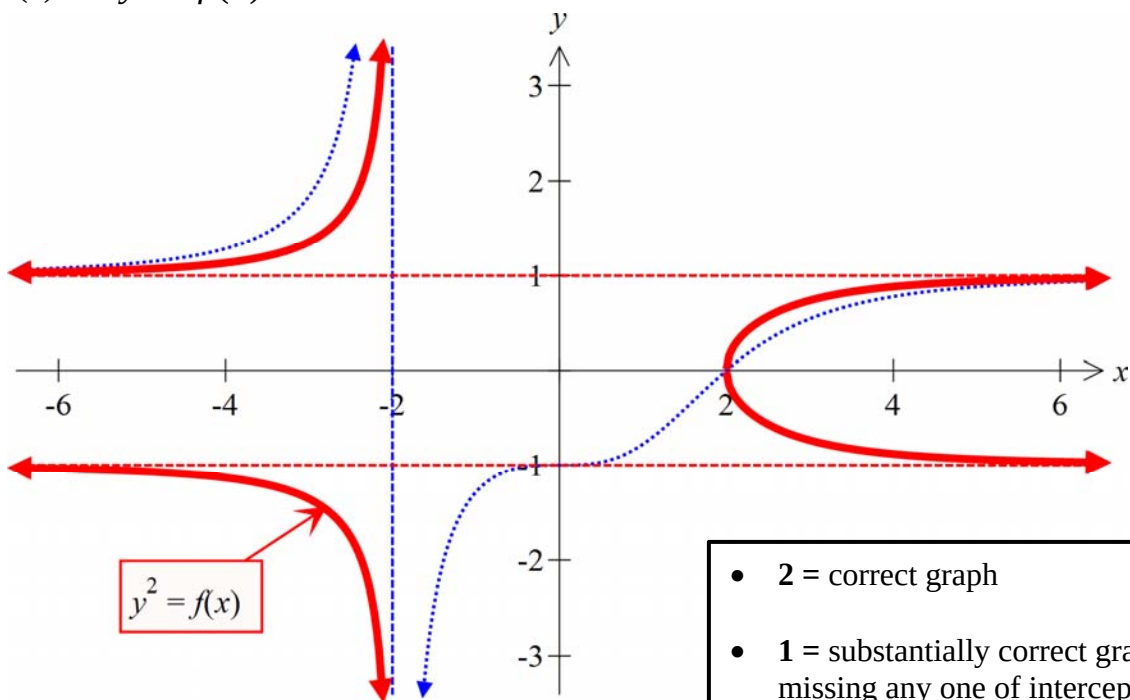
(iv) $|y| = f(x)$

1



(v) $y^2 = f(x)$

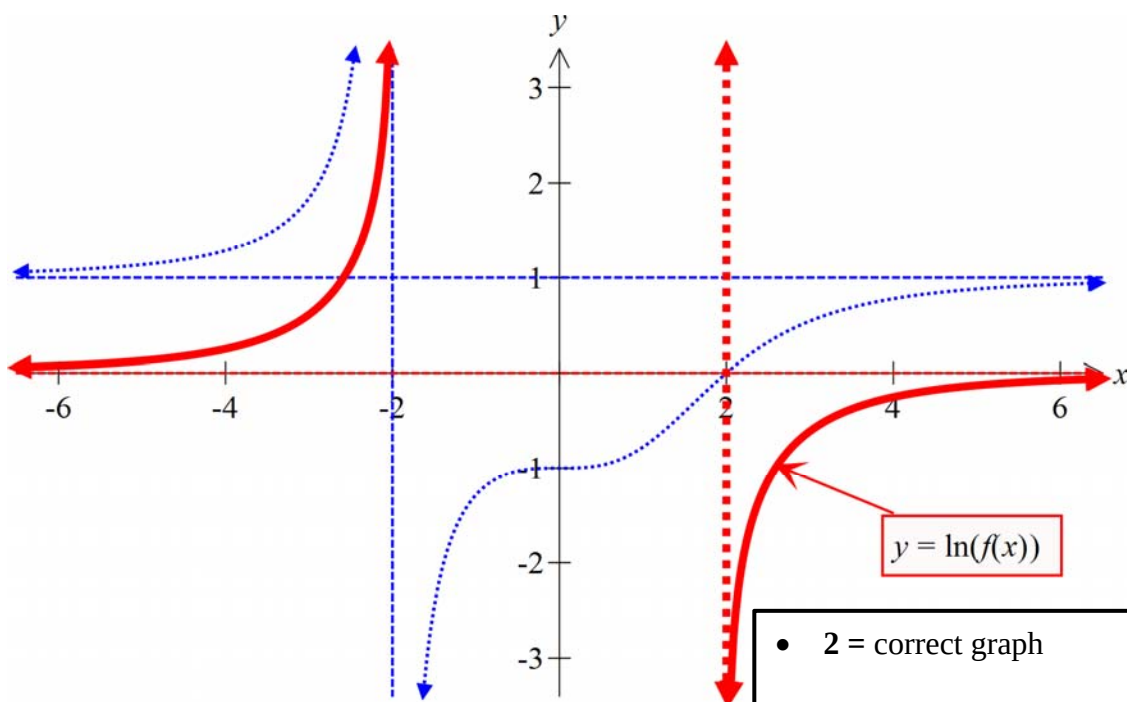
2



- 2 = correct graph
- 1 = substantially correct graph but missing any one of intercepts, asymptotes, behaviour at end points or intercepts, or stationary points or points of inflexion

(vi) $y = \log_e f(x)$

2



- 2 = correct graph
- 1 = substantially correct graph but missing any one of intercepts, asymptotes, behaviour at end points or intercepts, or stationary points or points of inflexion

(b) (ii)

Suggested solution	CRITERIA/COMMENTS	MARKS
Let $S(n)$ be the statement $3^n > n^3$	Correct solution including verification of the initial case	3
Step 1: Show $S(4)$ is true $LHS = 3^4 = 81$ $RHS = 3^3 = 27$ SINCE $81 > 27$ then $S(4)$ is true	Establishes the truth of $S(k+1)$ by correctly using the assumption $S(k)$ but ignores the initial case or equivalent method	2
Step 2: Assume $S(k)$ is true for $3 < k \leq n$, $(k, n) \in \mathbb{Z}^+$ i.e. $3^k > k^3$ (1) Aim to prove $S(k+1)$ is true i.e. $3^{k+1} > (k+1)^3$ (2) The statement (2) can be shown if we can show that for $k > 3$, $3^{k+1} - (k+1)^3 > 0$. $\begin{aligned} 3^{k+1} - (k+1)^3 &= 3(3^k) - (k+1)^3 \\ &> 3k^3 - (k+1)^3 \text{ from (1)} \\ &= 3 \times 4^3 - 5^3, \quad k = 4 \\ &= 67 \\ &> 0, \text{ if } k > 3 \end{aligned}$ Since $S(1)$ and if $S(k)$ is true then $S(k+1)$ is also true. So by the principle of mathematical induction $3^n > n^3$, for integral $n > 3$.	- Verifies at least the statement is true for $n = 4$ - Writes down a correct assumption e.g. Assume $3^k > k^3$ or $3^k - k^3 > 0$ No penalty for omission of a conclusion Other methods are available such as Calculus or consider: $3^k > k^3$ (assumption) $\therefore 3 \cdot 3^k > 3k^3$ $3^{k+1} > 3k^3$ Is $3k^3 > (k+1)^3$ Least value $k = 4$. $3 \times 64 > 4^3$ This suggests that $(k+1)^3 > 64$ (at least)	1

(c) (i)

Suggested solution	CRITERIA/COMMENTS	MARKS
$ \overline{OR} = 2 \overline{OP} $ $= 2\sqrt{\left(-\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}$ $= 2 \times 1$ $= 2$ $\arg(\overline{OP}) = 180^\circ - \tan^{-1}(\sqrt{3})$ $= 60^\circ$ $\therefore \arg(\overline{OR}) = (120 - 60)^\circ = 60^\circ$ i.e. $\overline{OR} = 2\text{cis}60^\circ$ $= 1 + \sqrt{3}i$	A correct solution in correct modulus and argument form or in correct Cartesian form	2
	- Numeric form of either one of $\overline{OP}$, $\overline{OR} = 2 \overline{OP}$, $\arg(\overline{OP})$, or $\arg(\overline{OR})$ - Recognising that $\overline{OR}$ is a rotation of $\overline{OP}$ clockwise by 60° but twice the modulus of $\overline{OP}$.	1

ALTERNATIVE: Given $\angle POR = 60^\circ$ To rotate $\overrightarrow{OP}$ clockwise by 60° we need to multiply $\overrightarrow{OP}$ by $\text{cis}(-60^\circ)$. i.e. $\overrightarrow{OR} = 2 \times \overrightarrow{OP} \times \text{cis}(-60^\circ)$ $= 2\text{cis}120^\circ \text{cis}(-60^\circ)$ $= 2\text{cis}60^\circ$		
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(c) (ii)

Suggested solution	CRITERIA/COMMENTS	MARKS
$\begin{aligned}\overrightarrow{OQ} &= \overrightarrow{OR} + \overrightarrow{OP} \\ &= 1 + \sqrt{3}i + \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) \\ &= \frac{1}{2} + \left(\sqrt{3} + \frac{\sqrt{3}}{2}\right)i \\ &\left(\cup \frac{1}{2} + \frac{3\sqrt{3}}{2}i\right)\end{aligned}$	Correct answer in Cartesian form only	1

QUESTION 14
MARKING GUIDELINES

(a) (i)

Suggested solution	CRITERIA/COMMENTS	MARKS
Want $x+1 \geq 0$ i.e. $D: x \geq -1$	Correct answer	1

(ii)

Suggested solution	CRITERIA/COMMENTS	MARKS
$Y - \text{INTERCEPT} : x = 0, y = -1$ $X - \text{INTERCEPT} : y = 0, x^2 - 1 = 0 \text{ or } x + 1 = 0$ i.e. $x = \pm 1$	Correct answers one y-intercept and two x-intercepts	2
	One x-intercept or y-intercept	1

(iii)

Suggested solution	CRITERIA/COMMENTS	MARKS
Given $\frac{dy}{dx} = \frac{5x^2 + 4x - 1}{2\sqrt{x+1}}$ and $\frac{d^2y}{dx^2} = \frac{3(5x^2 + 8x + 3)}{4(x+1)\sqrt{x+1}}$ SP's when $\frac{dy}{dx} = 0$ i.e. when $5x^2 + 4x - 1 = 0$ $(5x - 1)(x + 1) = 0$ $x = \frac{1}{5}$ or $x = -1$ When $x = \frac{1}{5}$, $y = -\frac{24\sqrt{30}}{125} (\approx -1.05)$ When $x = -1$, $y = 0$ NATURE USING y''. At $x = \frac{1}{5}$, $y'' = \frac{3(5(\frac{1}{5})^2 + 8(\frac{1}{5}) + 3)}{4(\frac{1}{5} + 1)\sqrt{\frac{1}{5} + 1}}$ > 0 $\therefore \left(\frac{1}{5}, -\frac{24\sqrt{30}}{125}\right)$ is a rel. min. SP At $x = -1$, y'' & y' are inconclusive tests. To establish what is happening at the point $(-1, 0)$, note that there is a possible point of inflexion when $5x^2 + 8x + 3 = 0$. That is when $(x + 1)(5x + 3) = 0$ or at $x = -1$ & $x = -\frac{3}{5}$. This suggests that the curve has greatest negative slope at $x = -\frac{3}{5}$, but the slope at $x = -1$ is undefined because it is an endpoint. Given $x \geq -1$, we suspect that the nature to the right of $(-1, 0)$ is a 'half' concave down feature to where $x = -\frac{3}{5}$. For $-1 < x < -\frac{3}{5}$, $y'' < 0$. Verify this by substitution of $x = -\frac{3}{5}$ into y''.	Correct solution/justification and establishing the nature and/or feature at $\left(\frac{1}{5}, -\frac{24\sqrt{30}}{125}\right)$ using Calculus or equivalent method	3
	- Correctly solves $\frac{dy}{dx} = 0$ giving the two x-values - Correctly finds the coordinates $\left(\frac{1}{5}, -\frac{24\sqrt{30}}{125}\right)$ or equivalent	2
	Correctly solves $\frac{dy}{dx} = 0$ giving one x-value (either $x = -\frac{1}{5}$ or $x = -1$)	1

(iv)

Suggested solution	CRITERIA/COMMENTS	MARKS
Point of Inflection (-0.6 , -0.405) Local Minimum (0.2 , -1.0516)	Correct graph showing all important features such as the behaviour at $x = -1$, intercepts and SP's	2
	Correct graph showing all important features but makes an error in at least one of behaviour at $x = -1$, intercepts, SP's or POI's	1

(v)

Suggested solution	CRITERIA/COMMENTS	MARKS
$A = \left \int_{-1}^1 (x^2 - 1)\sqrt{x+1} \, dx \right $ Let $u = x+1$, $\frac{du}{dx} = 1$, when $x = -1$, $u = 0$ when $x = 1$, $u = 2$ $\therefore A = \left \int_0^2 (u^2 - 2u)\sqrt{u} \, du \right $ $= \left \int_0^2 u^2\sqrt{u} - 2u\sqrt{u} \, du \right $ $= \left \int_0^2 u^{\frac{5}{2}} - 2u^{\frac{3}{2}} \, du \right $ $= \left \frac{2}{7}u^{\frac{7}{2}}\sqrt{u} - \frac{4}{5}u^{\frac{5}{2}}\sqrt{u} \right _0^2$ $= \left \frac{16}{7}\sqrt{2} - \frac{16}{5}\sqrt{2} \right $ $= \frac{32}{35}\sqrt{2} \text{ sq. units}$	Correct solution in the form $\frac{a\sqrt{b}}{c}$	4
	- A correct primitive of $(x^2 - 1)\sqrt{x+1}$, using a substitution such as $u = x+1$ or $u^2 = x+1$ or any appropriate method leading to the correct answer (if $u^2 = x+1$ is chosen then $x^2 - 1 = (u^2 - 1)^2 - 1$) - A correct primitive of their incorrect definite integral and an attempt to evaluate this definite integral but makes an error along the way (the integral expression must not be easier than intended, otherwise award a maximum of 2 marks)	3
	A correct definite integral expression with/without the absolute value sign, for example $A = \left \int_{-1}^1 (x^2 - 1)\sqrt{x+1} \, dx \right ,$ $\int_0^2 (u^2 - 2u)\sqrt{u} \, du$	2
	An attempt to use integration to find the area of the region specified	1

(b) (i)

Suggested solution	CRITERIA/COMMENTS	MARKS
$\cos(A+B) - \cos(A-B) = -2\sin A \sin B$	Correct answer	1

(ii)

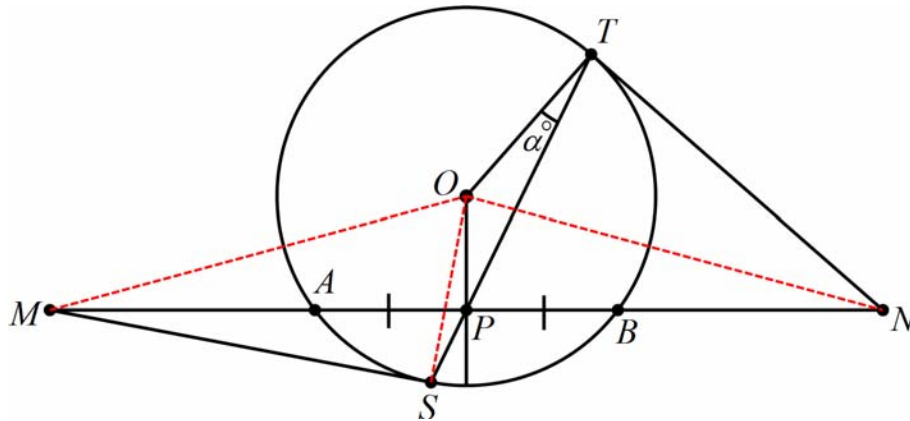
Suggested solution	CRITERIA/COMMENTS	MARKS
$\cos x - \cos 3x = -2\sin\left(\frac{x+3x}{2}\right)\sin\left(\frac{x-3x}{2}\right)$ $= -2\sin 2x \sin(-x)$ $= 2\sin 2x \sin x$ $\therefore 2\sin 2x \sin x = 0$ i.e. $\sin 2x = 0$ or $\sin x = 0$ $\therefore x = \frac{n\pi}{2} \quad x = n\pi$ Since the solution $x = n\pi$ is generated in the solution $x = \frac{n\pi}{2}$, for some integer value of n, then $x = \frac{n\pi}{2}$ only.	Correct solution using any appropriate method; answer must be in general form (but it is not unique)	2
	- Simplifying $\cos x - \cos 3x$ to $2\sin x \sin 2x$ or equivalent. - Stating without working $x = \frac{n\pi}{2}$ or $x = n\pi$ - An attempt to solve the given equation	1

QUESTION 15
MARKING GUIDELINES

(a)

Suggested solution	CRITERIA/COMMENTS	MARKS
Let $t = \tan \frac{\theta}{2}$, $\frac{dt}{d\theta} = \frac{1}{2} \sec^2 \frac{\theta}{2}$ & $d\theta = \frac{2dt}{1+t^2}$ Also, $\cos \theta = \frac{1-t^2}{1+t^2}$, $\sin \theta = \frac{2t}{1+t^2}$ $\therefore \int_0^{\frac{\pi}{2}} \frac{2}{1+\cos \theta + \sin \theta} d\theta$ $= \int_0^1 \frac{2}{1+\frac{1-t^2}{1+t^2} + \frac{2t}{1+t^2}} \frac{2}{1+t^2} dt$ $= 2 \int_0^1 \frac{1}{1+t} dt$ $= 2 \ln 1+t \Big _0^1$ $= 2 \ln 2 - 2 \ln 1$ $= 2 \ln 2 \text{ (or } \ln 4)$	Correct solution (must use the given substitution up to 2nd last step at least)	3
	Substantial evidence of using the substitution method on a correct integrand but made an error along the way (if limits are incorrect or if no change of limits or if answer is left as an indefinite answer award a maximum of 1 mark)	2
	- Correct change of limits or variable(s) or finding in terms of t, du, $\frac{d\theta}{dt}$ - Attempt to use the given substitution	1

(b) (i)



Construct OM , ON and OS as shown in the diagram above

Suggested solution	CRITERIA/COMMENTS	MARKS
$\angle OTN = 90^\circ$ (NT is a tangent to circle at T) $\angle OPN = 90^\circ$ (OP is $\perp$ bisector of chord AB) $\Rightarrow \angle OTN + \angle OPN = 180^\circ$ $\therefore OPNT$ is a cyclic quadrilateral (opposite angles of a quadrilateral are supp.) (or both are angles in a semi-circle on ON)	Correct solution including complete reasons	2
	$\angle OTN = 90^\circ$ or $\angle OPN = 90^\circ$ without reasons $\angle MPO = \angle OTN = 90^\circ$ exterior angle of cyclic quad equals int. opp. angle	1

(ii)

Suggested solution	CRITERIA/COMMENTS	MARKS
Construct OM and OS $\angle OSM = 90^\circ$ (MS is a tangent to circle at S) $\angle OPM = 90^\circ$ (OP is $\perp$ bisector of chord AB) $\Rightarrow \angle OSM = \angle OPM$ (equal angles standing on same line OM) $\therefore OPSM$ is a cyclic quadrilateral (opposite angles of a quadrilateral are supp.) (i.e. OM is a diameter)	Correct solution including complete reasons	2
	$\angle OSM = 90^\circ$ or $\angle OPM = 90^\circ$ without reasons	1

(iii)

Suggested solution	CRITERIA/COMMENTS	MARKS
$OPNT$ is a cyclic quadrilateral $\angle ONP = \angle OTP = \alpha$ (angles on same chord OP) $\angle OSP = \angle OTS = \alpha$ (base angles in isosceles $\triangle OST$) $OPSM$ is a cyclic quadrilateral $\therefore \angle OMP = \angle OSP = \alpha$ (angles on same chord OP) $\therefore \angle OMP = \angle ONP$	Correct solution including complete reasons (must refer to the cyclic quads. established in in (i) and (ii) in some way and a link made) Paraphrased reasons or in-concise reasons are unacceptable.	2
	Noting at least one other angle equivalent to $\angle OTS$ with or without reasons	1

(iv)

Suggested solution	CRITERIA/COMMENTS	MARKS
$\triangle OMN$ is isosceles (base angles equal) OP is an altitude of $\triangle OMN$ $\therefore MP = PN$ $AM = MP - AP$ $= PN - PB$ $= BN$	- Correct solution including complete reasons - Proving $\triangle OPN \equiv \triangle OPM$	2
	OP is an altitude of isosceles triangle OMN	1

(c) (i)

Suggested solution	CRITERIA/COMMENTS	MARKS
$f(x) = 2 \cos^{-1} \frac{x}{\sqrt{2}} - \sin^{-1}(1-x^2)$ $f'(x) = -2 \left[\frac{\frac{d}{dx} \left(\frac{x}{\sqrt{2}} \right)}{\sqrt{1 - \left(\frac{x}{\sqrt{2}} \right)^2}} \right] - \frac{\frac{d}{dx}(1-x^2)}{\sqrt{1 - (1-x^2)^2}}$ $= \frac{-2 \left(\frac{1}{\sqrt{2}} \right)}{\frac{1}{\sqrt{2}} \sqrt{2-x^2}} + \frac{2x}{\sqrt{1 - (1-2x^2+x^4)}}$ $= -\frac{2}{\sqrt{2-x^2}} + \frac{2x}{\sqrt{2x^2-x^4}}$ $= -\frac{2}{\sqrt{2-x^2}} + \frac{2x}{x\sqrt{2-x^2}}$ $= 0$	Correct simplified derivatives of $\cos^{-1} \frac{x}{\sqrt{2}}$ and $\sin^{-1}(1-x^2)$	2
	Correct derivative of either $\cos^{-1} \frac{x}{\sqrt{2}}$ or $\sin^{-1}(1-x^2)$ unsimplified	1

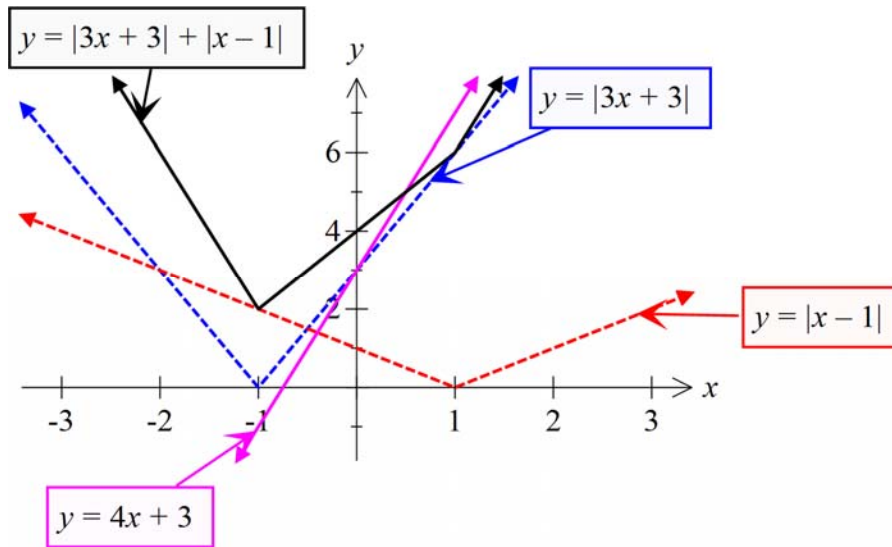
(ii)

Suggested solution	CRITERIA/COMMENTS	MARKS
$f'(x) = 0, \therefore f(x)$ is a constant function. For $0 \leq x \leq 1$, let $x = 1$, $f(1) = 2 \cos^{-1} \frac{1}{\sqrt{2}} - \sin^{-1}(1-1)$ $= \frac{\pi}{2}$ $\therefore \int_0^1 f(x) dx = \int_0^1 \frac{\pi}{2} dx$ $= \left[\frac{\pi x}{2} \right]_0^1$ $= \frac{\pi}{2}$	- Correct solution (may or may not include a graph) but including establishment that $f(x) = \frac{\pi}{2}$, in $0 \leq x \leq 1$ - Integration by parts is acceptable but not necessary.	2
	- Explanation that $f(x)$ is a constant function $f(x) = \frac{\pi}{2}$, in $0 \leq x \leq 1$ $\int_0^1 f(x) dx = \int_0^1 \frac{\pi}{2} dx$	1

QUESTION 16
MARKING GUIDELINES

(a)

(i)



3 = correctly drawn graphs of $y = |3x + 3| + |x - 1|$ & $y = 4x + 3$

2 = correctly drawn graphs of $y = 4x + 3$ and $y = |3x + 3|$ or $y = |x - 1|$

1 = correctly drawn graphs of $y = 4x + 3$ or $y = |3x + 3|$ or $y = |x - 1|$

(ii)

Suggested solution	CRITERIA/COMMENTS	MARKS
correctly drawn graphs of $3x + 3 + x - 1 \leq 4x + 3$ Consider the intersection of $y = 4x + 3$ and $y = (3x + 3) - (x - 1) \Rightarrow$ see graph $= 2x + 4$ Solving $2x + 4 = 4x + 3$ $x = \frac{1}{2}$ $\therefore x \geq \frac{1}{2}$ only	Correct solution from their graphs in (i) provided they are well labelled and distinguished from one another and the problem has not been made easier than intended	2
	- Establishing the graphs of $y = 4x + 3$ and $y = (3x + 3) - (x - 1)$ intersect at $x = \frac{1}{2}$. - Solving appropriate equations for the following cases, $x \geq 1, -1 \leq x \leq 1, x < -1$	1

(b) (i)

Suggested solution	CRITERIA/COMMENTS	MARKS
$I_0 = \int_0^1 e^{-x} dx = -[e^{-x}]_0^1 = 1 - \frac{1}{e}$	Correct answer only	1

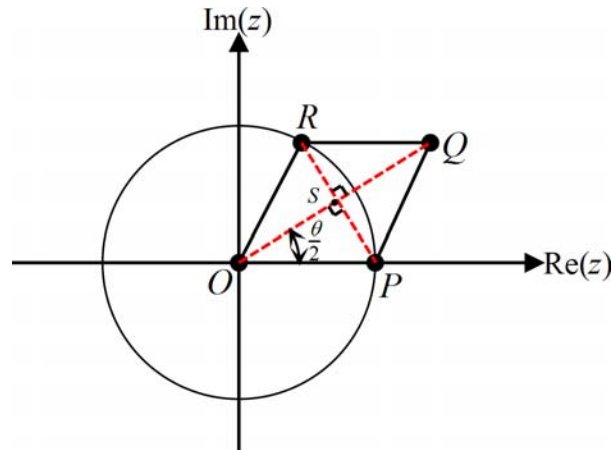
(ii)

Suggested solution	CRITERIA/COMMENTS	MARKS
$\text{Let } I_n = \int_0^1 x^n e^{-x} dx$ $\text{Let } u = x^n \quad dv = e^{-x}$ $du = nx^{n-1} \quad v = -e^{-x}$ $I_n = uv - \int v du$ $= -\left[x^n e^{-x}\right]_0^1 + n \int_0^1 x^{n-1} e^{-x} dx$ $\text{i.e. } I_n = -[e^{-1}] + nI_{n-1}$ $= nI_{n-1} - \frac{1}{e}$	Correct proof	2
	- Correct primitive of their dv - Correct derivative of their u - Correct evaluation of either $-\left[x^n e^{-x}\right]_0^1$ or $n \int_0^1 x^{n-1} e^{-x} dx$	1

(iii)

Suggested solution	CRITERIA/COMMENTS	MARKS
$\text{Let } I_4 = \int_0^1 x^4 e^{-x} dx$ $I_4 = 4I_3 - \frac{1}{e}$ $= 4\left[3I_2 - \frac{1}{e}\right] - \frac{1}{e}$ $= 4\left[3\left(2I_1 - \frac{1}{e}\right) - \frac{1}{e}\right] - \frac{1}{e}$ $= 4\left[3\left(2\left\{I_0 - \frac{1}{e}\right\} - \frac{1}{e}\right) - \frac{1}{e}\right] - \frac{1}{e}$ $= 4\left[3\left(\left\{2I_0 - \frac{2}{e}\right\} - \frac{1}{e}\right) - \frac{1}{e}\right] - \frac{1}{e}$ $= 4\left[\left(\left\{6I_0 - \frac{6}{e}\right\} - \frac{3}{e}\right) - \frac{1}{e}\right] - \frac{1}{e}$ $= 24I_0 - \frac{24}{e} - \frac{12}{e} - \frac{4}{e} - \frac{1}{e}$ $= 24\left(1 - \frac{1}{e}\right) - \frac{41}{e}$ $= 24 - \frac{65}{e}$	Correct solution	2
	- Statement $I_4 = \int_0^1 x^4 e^{-x} dx$ - Statement $I_4 = 4I_3 - \frac{1}{e}$	1

(c) (i)



Construct diagonals OQ and PR of rhombus $OPQR$. They intersect at right-angles (property of a rhombus) at S (see diagram)

Suggested solution	CRITERIA/COMMENTS	MARKS
$\angle QOP = \frac{\theta}{2}$ (diagonals of a rhombus bisect the angles through which they pass) $\therefore z_2 = \overrightarrow{OQ} = \overrightarrow{OQ} \operatorname{cis} \frac{\theta}{2}$ But $ \overrightarrow{OQ} = 2 \overrightarrow{OS} $ & $ \overrightarrow{OP} = 1$ (given) In right $\triangle SOP$, $\frac{ \overrightarrow{OS} }{ \overrightarrow{OP} } = \cos \frac{\theta}{2}$ $\therefore \overrightarrow{OS} = \cos \frac{\theta}{2}$ & $ \overrightarrow{OQ} = 2 \cos \frac{\theta}{2}$ Hence, $z_2 = 2 \cos \frac{\theta}{2} \operatorname{cis} \frac{\theta}{2}$ $= 2 \cos \frac{\theta}{2} \left[\cos \frac{\theta}{2} + i \sin \frac{\theta}{2} \right]$	Correct proof including all three of $\angle QOP = \frac{\theta}{2}$ & $z_2 = \overrightarrow{OQ} = \overrightarrow{OQ} \operatorname{cis} \frac{\theta}{2}$ & $\overrightarrow{OS} = \cos \frac{\theta}{2}$ or equivalent notation and fact(s)	3
$\therefore \overrightarrow{OS} = \cos \frac{\theta}{2}$ & $ \overrightarrow{OQ} = 2 \cos \frac{\theta}{2}$ Hence, $z_2 = 2 \cos \frac{\theta}{2} \operatorname{cis} \frac{\theta}{2}$ $= 2 \cos \frac{\theta}{2} \left[\cos \frac{\theta}{2} + i \sin \frac{\theta}{2} \right]$	Identifying a minimum of two facts from $\angle QOP = \frac{\theta}{2}$ or $z_2 = \overrightarrow{OQ} = \overrightarrow{OQ} \operatorname{cis} \frac{\theta}{2}$ or $\overrightarrow{OS} = \cos \frac{\theta}{2}$ or equivalent notation and fact(s)	2
Can use the facts that $1 + \cos \theta = 2 \cos^2 \left(\frac{\theta}{2} \right)$ $\sin \theta = 2 \sin \left(\frac{\theta}{2} \right) \cos \left(\frac{\theta}{2} \right)$ and just simplify the given expression	Identifying only one of $\angle QOP = \frac{\theta}{2}$ or $z_2 = \overrightarrow{OQ} = \overrightarrow{OQ} \operatorname{cis} \frac{\theta}{2}$ or $\overrightarrow{OS} = \cos \frac{\theta}{2}$ or equivalent notation and fact(s)	1

(ii)

Suggested solution	CRITERIA/COMMENTS	MARKS
From (i) $z_2 = 2 \cos \frac{\theta}{2} \operatorname{cis} \frac{\theta}{2}$ $\therefore \frac{1}{z_2} = \frac{\left[\cos \frac{\theta}{2} + i \sin \frac{\theta}{2} \right]^{-1}}{2 \cos \frac{\theta}{2}}$ $= \frac{\left[\cos \frac{\theta}{2} - i \sin \frac{\theta}{2} \right]}{2 \cos \frac{\theta}{2}} \text{ (de Moivre's Th)}$ $= \frac{1}{2} \left[\frac{\cos \frac{\theta}{2}}{\cos \frac{\theta}{2}} - i \frac{\sin \frac{\theta}{2}}{\cos \frac{\theta}{2}} \right]$ $= \frac{1}{2} - \frac{i}{2} \tan \frac{\theta}{2}$	- Correct proof including a clear indication of the splitting of terms (see second last line of suggested solution) - Can realise the denominator	2
	Statement $\frac{1}{z_2} = \frac{\left[\cos \frac{\theta}{2} + i \sin \frac{\theta}{2} \right]^{-1}}{2 \cos \frac{\theta}{2}} \text{ or}$ $\frac{\left[\cos \frac{\theta}{2} - i \sin \frac{\theta}{2} \right]}{2 \cos \frac{\theta}{2}} \text{ from de Moivre's Theorem}$	1

End of Marking Guidelines, Criteria and Solutions